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# Experiment and numerical investigations of ultra-high toughness cementitious composite slabs under contact explosions

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ABSTRACT

Reinforced concrete (RC) structures under blast loading can occur severe structural damage, including cratering, spalling, and breaching. And these damages could cause danger to the personnel and equipment. In this study, ultra-high toughness cementitious composite (UHTCC) with significant strain-hardening behavior (ultimate tensile strain >3%) was utilized to improve the blast resistance of construction structure. The UHTCC square slabs with a side length of 2000 mm were tested under contact explosions. The cratering and spalling damage of slabs were investigated. A total of six slabs (three RC slabs and three UHTCC slabs) under three different weight of TNT charge (2 kg, 4 kg, and 5 kg) were tested. The results showed the UHTCC slabs presented excellent blast resistance compared with RC slabs. The existing empirical predictions were employed to explore the damage mode of UHTCC slabs compared with the experimental results. The suitability and accuracy of the predictions were discussed. The structured and unstructured discretizations with different mesh size were used in the finite element (FE) model. The simulation results from the unstructured discretizations show better agreement with the experimental results than that from the structured discretizations. The effect of mesh size on the damaged area, damaged depth, deflection-time curves of the slab and the eroded volume fraction were compared and discussed. In addition, the sensitivity analysis of input parameters in the numerical model were conducted. the maximum failure strain, the compressive strength and the mass density of UHTCC were employed as the input parameters to investigate and discuss their effects on the value of output parameters, which include the damaged area, damaged depth and maximum deflection of UHTCC slab.

## 1. Introduction

The growing number of terrorist attacks around the world have taken their toll on human lives and caused economic loss. The explosive attack, which includes the contact explosion and non-contact explosion, is a common type of terrorist attack. When subjected to explosive loadings, the concrete structures may present local damage, including cratering, spalling and breaching damage. As shown in Fig. 1, cratering damage occurs because of high compressive pressure in the contacted surface between the concrete and charges. Spalling damage appears in the free surface of the concrete structures when the tensile stress generated by the compressive waves in the free surface is larger than the spalling strength of the concrete. Breaching damage arises when the impulse generated by the explosion is large enough to overcome the resistant forces of structures [1].

In the blast event, the high-speed fragments produced by the spalling and breaching damage of structures could cause unexpected casualties and property loss [3–5]. Therefore, it is most significant to prevent and decrease the spalling and breaching damage of the structures under blast loadings.

To solve this problem, Morishita et al. [6], as well as Tanaka and Tsuji [7] attempted to increase the reinforcement ratio and compressive strength of concrete to obtain an outstanding blast resistance of concrete structures, but the results of the tests were not consistent with the desired effect. In the protective structures against blast loadings, Nam et al. [8] found that the energy absorption capacity and the fracture energy of materials have an obvious influence on the blast resistance performance of structures. In the current construction materials, ultra-high toughness cementitious composite (UHTCC) shows the excellent energy absorption ability with ultimate tensile strain over 3.0% [9–13], And it also shows excellent fracture energy with 5371 N/m [14], which is 32 times more than ordinary concrete with 167 N/m [15]. That indicates UHTCC can be employed to prevent and decrease spalling damage. Furthermore, UHTCC presents better durability [16], fire performance [17,18], the excellent cyclic [19], fatigue [20–22] and impact performance [11,23–25]. Because of those advantages and novel constructive technologies of UHTCC [26–29], UHTCC could have great

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**Nomenclature**

*Parameters for empirical prediction*

$K_b$[m/kg$^{1/3}$] blast-resistant coefficient
$h$[m] damaged depth of cratering
$W$[kg] weight of TNT
$R$[m] standoff distance
$\rho_c$[kg/m$^3$] mass density of TNT
$d$[m] diameter of TNT
H[m] height of TNT
$T$[m] thickness of the slab
$R/W^{1/3}$[m/kg$^{1/3}$] scaled distance
$T/W^{1/3}$[m/kg$^{1/3}$] scald thickness

*Symbols and parameters for K&C model*

$\rho_u$[kg/m$^3$] mass density of UHTCC
$\mathbf{u}$ displacement vector
$\boldsymbol{\sigma}$ Cauchy stress tensor
$\nabla_x^s$ symmetric spatial gradient operator
$\mathbf{b}$ body force
$\mathbf{t}$ traction on the natural boundary
$\partial\Omega_x^t$ natural boundary
$\Omega_x$ problem domain
$F(p,\boldsymbol{\sigma},\lambda)$ strength surface
$\boldsymbol{\sigma}'$ deviatoric stress tensor
$J_2$ the second invariant of $\boldsymbol{\sigma}'$
$J_3$ The third invariant of $\boldsymbol{\sigma}'$
$p$[MPa] hydrostatic pressure
$\lambda$[-] damage parameter
$r_f$[-] dynamic increase factor
$\eta(\lambda)$ damage evolution function
$\Delta\sigma_m$[MPa] maximum strength surface
$\Delta\sigma_r$[MPa] residual strength surface
$\Delta\sigma_y$[MPa] yield strength surface
$a_0$[MPa] parameter of $\Delta\sigma_m$
$a_1$[-] parameter of $\Delta\sigma_m$
$a_2$[MPa$^{-1}$] parameter of $\Delta\sigma_m$
$a_{0y}$[MPa] parameter of $\Delta\sigma_y$
$a_{1y}$[-] parameter of $\Delta\sigma_y$
$a_{2y}$[MPa$^{-1}$] parameter of $\Delta\sigma_y$
$a_{0f}$[MPa] parameter of $\Delta\sigma_y$
$a_{1f}$[-] parameter of $\Delta\sigma_y$
$a_{2f}$[MPa$^{-1}$] parameter of $\Delta\sigma_y$
$\overline{d\varepsilon^p}$ effective plastic strain increment
$d\varepsilon_{ij}^p$ plastic strain increment
$b_1$[-] parameter of $\eta(\lambda)$
$b_2$[-] parameter of $\eta(\lambda)$
$C(\mu)$[MPa] tabulated pressure parameter
$\theta(\mu)$[-] tabulated temperature parameter
$\gamma_0$[-] parameter of EOS-8
$E_0$[GPa] initial elastic modulus in EOS-8
$\mu$[-] volume strain in EOS-8
$K_v$[GPa] volume modulus in EOS-8
$r_{ft}$[-] $r_f$ for tension
$r_{fc}$[-] $r_f$ for compression
$F_{mt}$[-] parameter of $r_{ft}$
$W_{xt}$[-] parameter of $r_{ft}$
$S_t$[-] parameter of $r_{ft}$
$W_{yt}$[-] parameter of $r_{ft}$
$F_{mc}$[-] parameter of $r_{fc}$
$W_{xc}$[-] parameter of $r_{fc}$
$S_c$[-] parameter of $r_{fc}$
$W_{yc}$[-] parameter of $r_{fc}$
$\dot{\varepsilon}_u$[s$^{-1}$] strain rate of UHTCC
$\dot{\varepsilon}_0$[s$^{-1}$] quasi-state strain rate of UHTCC

*Parameters for air and TNT model*

$\sigma_{ys}$[MPa] yield strength of steel
$\sigma_0$[MPa] initial yield strength of steel
$E_p$[GPa] plastic hardening modulus of steel
$\varepsilon_{eff}^p$[-] effective plastic strain of steel
$C$[-] strain rate parameter of steel
$\gamma$[-] constant of EOS-1
$\dot{\varepsilon}$[s$^{-1}$] strain rate of steel
$\beta$[-] parameter of MAT_03
$\rho_a$[kg/m$^3$] mass density of air
$e_a$[kJ/kg] specific internal energy of air
$A$[MPa] parameter of JWL equation
$B$[MPa] parameter of JWL equation
$R_1$[-] parameter of JWL equation
$R_2$[-] parameter of JWL equation
$V$[m$^3$] specific volumes of TNT
$E$[kJ/kg] internal internal energy of TNT
$\omega$[-] parameter of JWL equation

*Symbols and parameters for uncertain analysis*

$\mathbf{X}$ input parameters vector
$X_1,\ldots,X_n$ input parameters
$\Psi_j$ orthogonal with respect to the Gaussian measure
$Z_j$ PCE deterministic coefficients
$\boldsymbol{\xi}$ vector of independent standard normal (Gaussian) random
$N$ polynomial of order
$n$ the number of imput parameters
$\boldsymbol{Y_s}$ vector of output parameters
$Y_1,\ldots,Y_n$ output parameters
S sum of the squares of the residuals
$\boldsymbol{Y_{as}}$ approximated solution of PCE
$R^2$ coefficient of determination
$R^2_{adj}$ adjust coefficient of determination
$f_{pc}(x)$ an integrable function
$k$ dimentions
$S_i$ first order indices
$S_{-i}$ the sum of all $S_{i_1,\ldots,i_s}$ except index $i$
$S_{Ti}$ Sensitivity indices
$E$ [-] mathematical expectation
$f_c$[MPa] compressive strength of UHTCC
$\varepsilon_{fmax}$[-] maximum failure strain

prospects in the engineering structures against blast loadings.

It is regretted that only few researches were conducted on the spalling resistant performance of UHTCC, and only a small amount of TNT (≤1kg) was applied to in the existing blast tests on UHTCC. Therefore, the blast experiments for UHTCC under a larger weight of TNT need to be studied. The primary purpose of conducting the blast tests of UHTCC is to provide adequate structural protection against explosions. As shown in Fig. 2, based on the excellent energy absorption property and high fracture energy of UHTCC, it is promising to prevent and decrease the spalling damage to the personnel. Moreover, the test results can also provide data for calibrating and verifying the dynamic constitutive model of UHTCC. In this study, Three UHTCC slabs were prepared, and three ordinary concrete slabs were prepared as a control group. 2 kg, 4 kg and 5 kg TNT explosives were employed to generate the blast loadings.

In addition to the experimental approach, the numerical analysis is a

good option to analysis the dynamic response of UHTCC subjected to blast loadings. The numerical results depend on the computational methods and the constitutive model of materials. The computational methods include discrete and continuous approaches. When subjected to blast loadings, the multi-cracks may appear in UHTCC members. In the past decades, some innovative and sophisticated numerical methods for simulating brittle cracking have been proposed and presented, such as particle-based methods [30–32], Dual-horizon peridynamics (DH-PD) [33,34], non-uniform horizon (NHPD) [35], cracking element methods (CEMs) [36–38], phase field methods (PFM) [39] and remeshing and interface elements. The cracking particle method, proposed by Rabczuk et al. [30–32], is a novel approach to model the discrete cracks in meshfree method, where the regions of cracking is represented by employing the piece-wise disconnected cracking segments. This method can significantly simplifying the crack tracking process. Dual-horizon peridynamics (DH-PD) [33,34] and NHPD [35] can distinctly deal with the crack stability problems on random particle arrangement. When solving the fluid-structure interaction of structures subjected to impulsive loadings, Rabczuk [40] proposed a significative coupling method which can solve the complicated cracking and perforation problems. In addition, the unstructured discretizations were also proved to provide better results, and demonstrate the superior accuracy on the simulation of fluid/structure interaction problems [41–44]. At present, the unstructured discretizations were rarely used in the blast simulations. In this study, both the unstructured and structured discretizations are employed in the finite models, and the relevant computational results from the two types of discretizations are compared and discussed.

In the numerical simulations, various uncertain input parameters could affect the numerical results of UHTCC when subjected to blast loadings. Therefore, the sensitive analysis (SA) for parameters is essential to obtain the sensitivity of parameters. Generally, number of model evaluations are essential to obtain a suitable sensitivity index. However, at least thousands of model evaluations must be conducted, which is expensive and utilizing several minutes to hours of CPU usage to simulate the dynamic response of UHTCC. Hence the SA for the full data is not recommended. At present, the surrogate model is popular for uncertain analysis. The surrogate models, can obtain a new computational model to provide a time-saving analysis based on a several number of output values generated from the original computational model. The surrogate models are popular for the rational treatment of uncertainties. Among the current surrogate models, the polynomial chaos expansions (PCE) model [45–47] is widely employed to study the random response of uncertain input parameters to the computational model by using the orthogonal polynomials in the random input parameters, such as the flexoelectric nanostructures [48], computational fracture models [49,50] and so on. PCE could also be employed as the stochastic finite element methods aiming at representing in an intrinsic way the complete response probability density function (PDF) [51]. The sobol indices can also be calculated based on the PCE [46,47].

Following the SA method used in Ref. [50], this study also employed

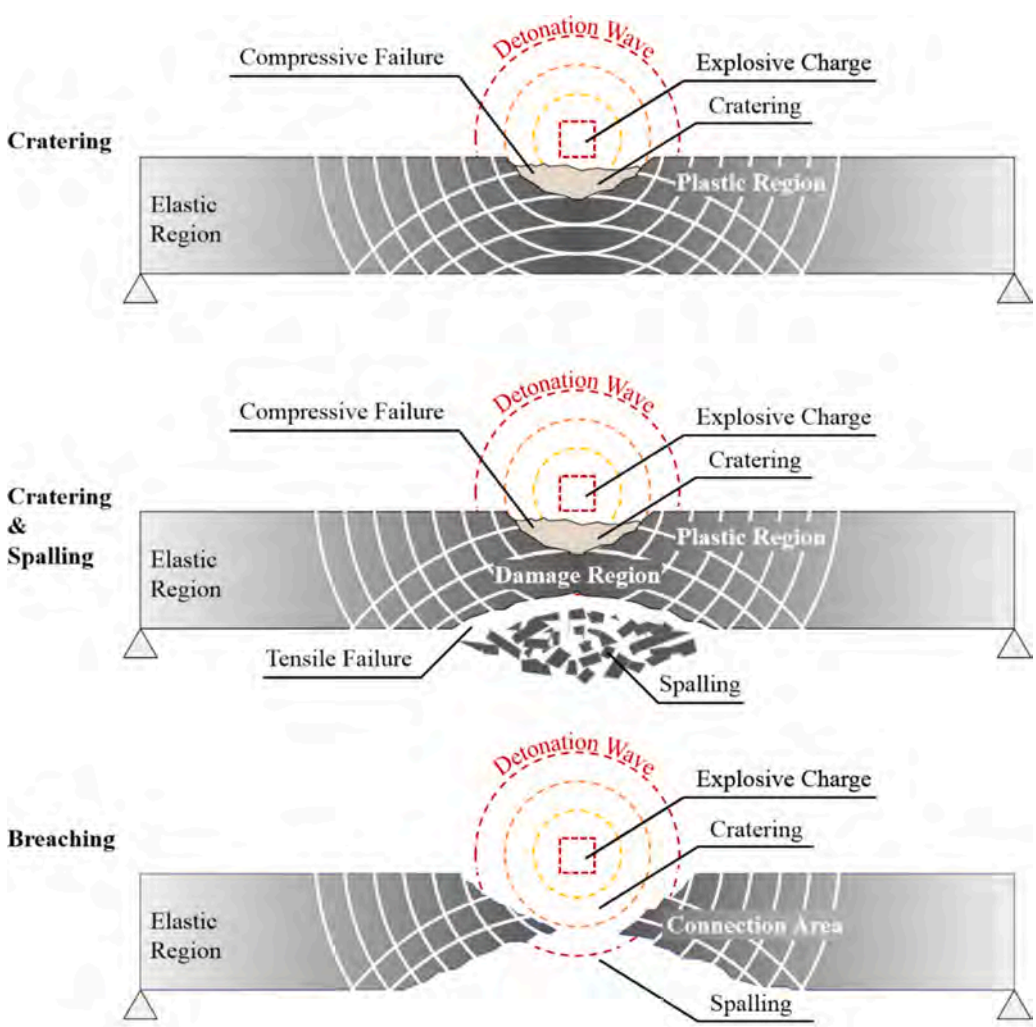


**Fig. 1.** Cratering, spalling and breaching damage of concrete structures [2].

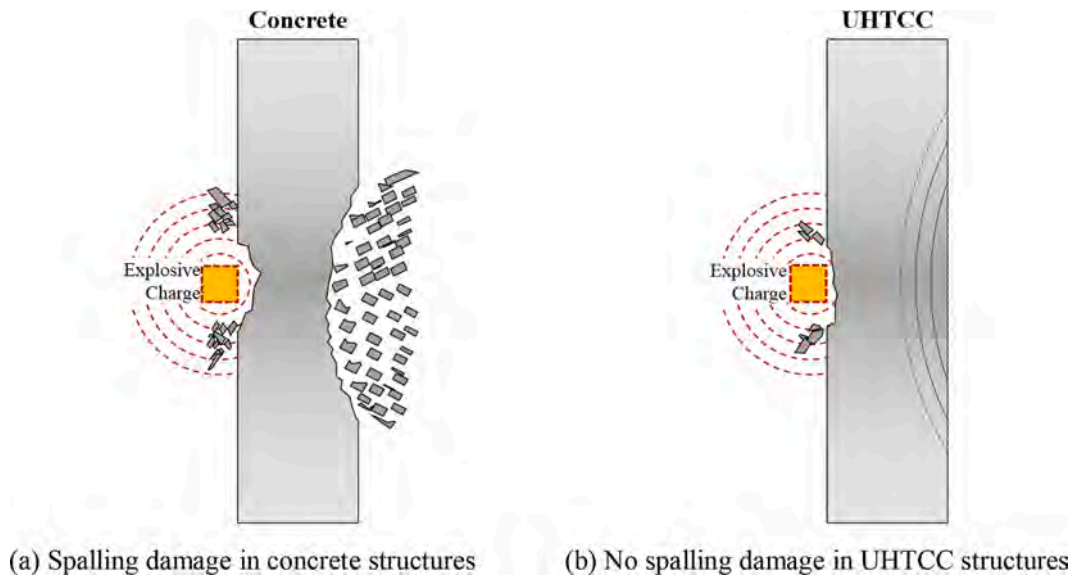


**Fig. 2.** Use UHTCC to decrease the spalling damage.

the PCE to conduct SA to quantify the influence of uncertain input parameters of the numerical model on the output parameters we investigate. The uncertain input parameters in the numerical simulation model include the compressive strength, mass density, and the maximum falilure strain. The output parameters include the damaed area, and the maximum deflection of the UHTCC slab under blast loadings. The PCE model could be established between the input and output parameters and the sensitivity inducis will be obtained further.

The content of the paper is outlined as follows. Section 2 shows the blast experimental preparation. Section 3 presents the experimental results and relevant discussions. Section 4 employs the current empirical formulations to predict the failure modes of UHTCC slabs and the feasibility of formulaes were discussed. Section 5 employed the structured and unctructured discretizations to conduct the numerical simulations, and the difference of structured and unstructured discretizations will be compared and discussed. The effect of mesh size on the damaged area, damaged depth, deflection-time curves of the slab and the eroded volume fraction were compared and discussed. In Section 6, the uncertainty analysis for the input parameters is conducted. The PCE and the parameters sensitivity are discussed. Section 7 closes the study with concluding remarks and future works.

## 2. Experimental preparation

### 2.1. Preparation of materials and specimens

UHTCC [52] consisted of ordinary Portland cement P•O 52.5, ultra-fine sand with an average size of 110 μm, ASTM class-F fly ash (FA), water, and the superplasticizer (SP) with a water-reducing rate of more than 35%. And 2% volume fraction of the K-II REC15 PVA fiber was used in UHTCC. The parameters of PVA fiber are listed in Table 1, and the appearance of PVA fibers is shown in Fig. 3.

In the casting process, the Portland cement P•O 52.5, sand, FA, and SP were poured into the blender and were dry-mixed for about 5 min at first. Then the water was added and mixed for about 5 min. After that, the PVA fibers were dispersed into the blender and were mixed for about 10 min with the slurry. Finally, the UHTCC slurry was prepared.

**Fig. 3.** PVA fibers

As shown in Fig. 4, before casting the slabs, the plastic films were fixed inside the mold for better curing. Then the steel bars were put into the mold. After that, poured the mixed UHTCC slurry into the molds,

**Fig. 4.** Process of preparing slabs.

**Table 1**
Parameters of PVA fiber.

| Fiber type | Diameter (μm) | Length (mm) | Tensile strength (MPa) | Elongation (%) | Young's modulus (GPa) | Density (g/cm$^3$) |
|---|---|---|---|---|---|---|
| PVA | 39 | 12 | 1600 | 6 | 40 | 1.3 |

vibrated for about 90 s and plastered the UHTCC slabs. After casting, slabs were covered with plastic sheeting immediately. One day later, all specimens were coved with wet fabrics and cured for 28 days.

The dimensions of the wooden mold were 2000 mm long by 2000 mm wide by 300 mm and 400 mm deep. Fig. 5 shows the dimensions and reinforcement layouts of slabs. The reinforcement was made into two layers, and the thickness of the protective layer was 30 mm. The HRB400 ribbed steel reinforcements were utilized as longitudinal reinforcements with a spacing of 100 mm, the HPB300 plain steel reinforcements with a normalized diameter of 8 mm were utilized as tie reinforcements and arranged in quincunx. In addition, three C40 concrete slabs were prepared as a control group. The mix proportion of C40 concrete is listed in Table 2.

### 2.2. Mechanical properties of the material

The compressive strength of UHTCC was obtained by using cubes with a dimension of 70.7 mm × 70.7 mm × 70.7 mm according to the Chinese Standard JGJ/T 70-2009 [53]. The tensile behavior of UHTCC was obtained by using dog-bone shaped specimens according to the recommendation of JSCE [54], and the dimension of specimens is shown in Fig. 6(a). According to the Chinese Standard JGJ/T 70-2009 [53], the prismatic specimens with dimensions of 100 mm × 100 mm × 400 mm were prepared for four-point bending experiments of concrete and UHTCC . All specimens for mechanical properties were cast and cured with the slabs. It keeps the slurry of specimens and slabs in the same condition. The load rate of tensile experiments was 0.2 mm/min, and that was 0.5 mm/min for compressive and four-point bending experiments. The typical tensile stress-strain curve was shown in Fig. 6(b). The compressive and tensile strength of UHTCC is 48 MPa and 4.5 MPa, respectively. The setup of four-point bending experiments and test results were shown in Fig. 7.

**Table 2**
Mix proportion of C40 concrete.

| Materials | Weight (kg/m$^3$) |
|---|---|
| Portland cement (P.O. 42.5) | 340 |
| Fine aggregate | 775 |
| Coarse aggregate | 1115 |
| Fly Ash (FA) | 60 |
| Water | 160 |
| Water reducing admixture | 4.4 |

### 2.3. Blast procedure

#### 2.3.1. Explosive charges

The TNT explosives with a heat of detonation of 4521 kJ/kg and a mass density of 1.57 g/cm$^3$ were employed in the tests. And the TNT explosives were made into cylinders with the same diameter (13 cm) and different heights. The weight of charges was 2 kg, 4 kg and 5 kg respectively. Fig. 8 shows the dimensions of TNT explosives. In addition, the electrical detonator was inserted and bonded together with the TNT. In the blast test, the charge was put on the central point of the slabs.

#### 2.3.2. Experimental setup

A total of 6 slabs were conducted in this study, including three concrete slabs and three UHTCC slabs, and the slabs were subjected to explosive loadings generated by 2 kg, 4 kg, and 5 kg TNT charge. 300 mm thick slabs were prepared for 2 kg and 4 kg TNT blast tests, and 400 mm thick slabs were prepared for 5 kg TNT blast tests. The test program is listed in Table 3. The boundary conditions with simply supported were adopted in the tests, and the width of the support boundary was 200 mm. A Laser Displacement Sensor (LDS) was employed to measure the deflection of the mid-span of slabs. The LDS had a stroke of up to 200

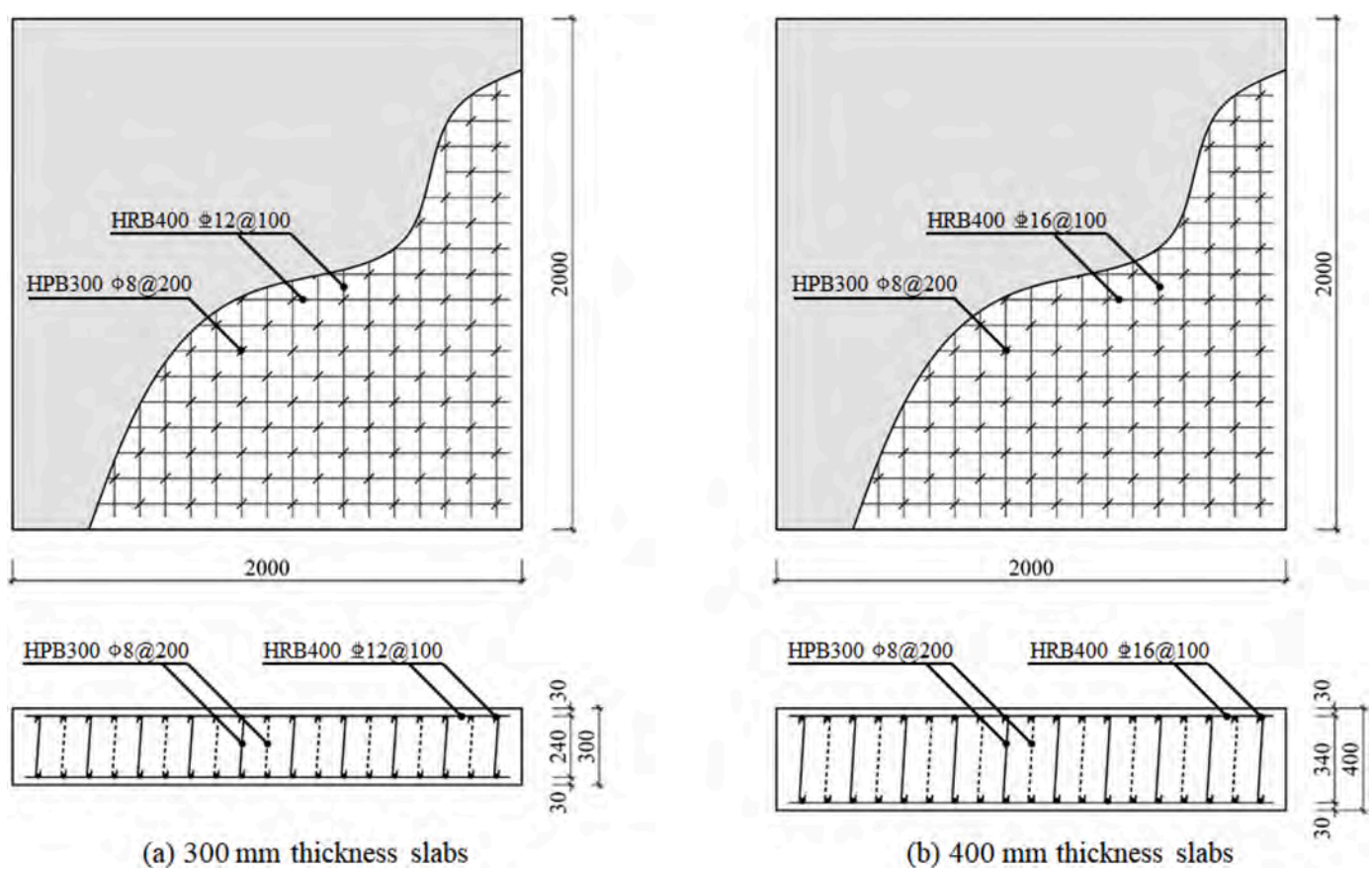


**Fig. 5.** The dimensions and reinforcement layouts of slabs.

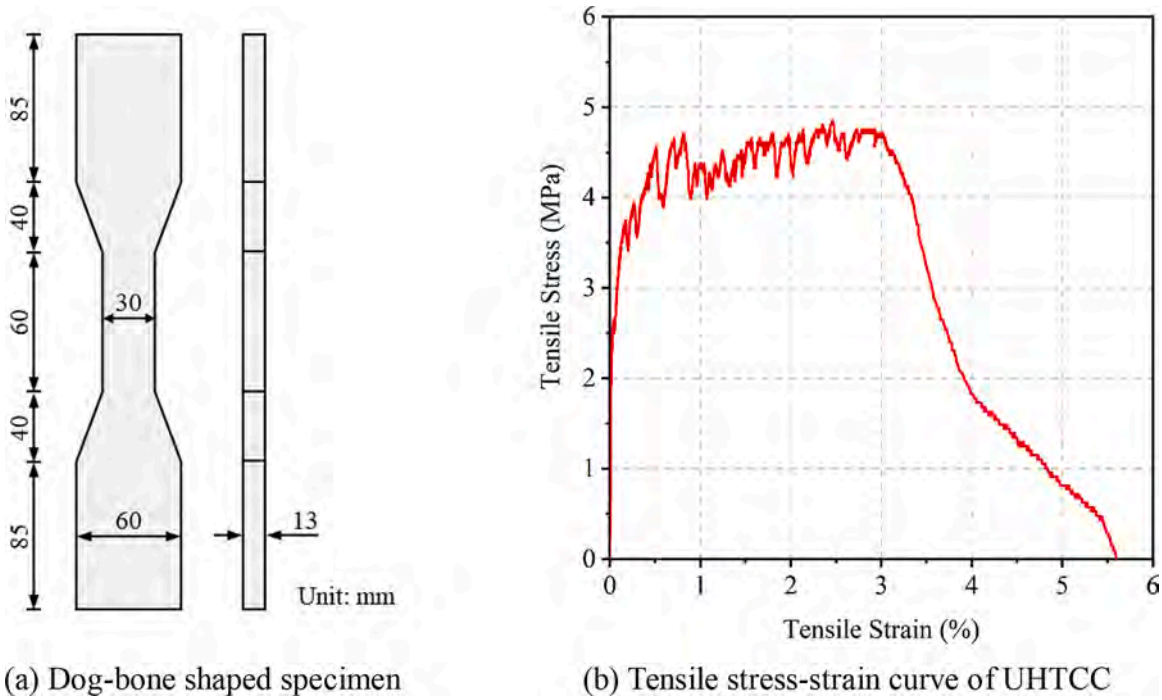


**Fig. 6.** Tensile test and results of UHTCC

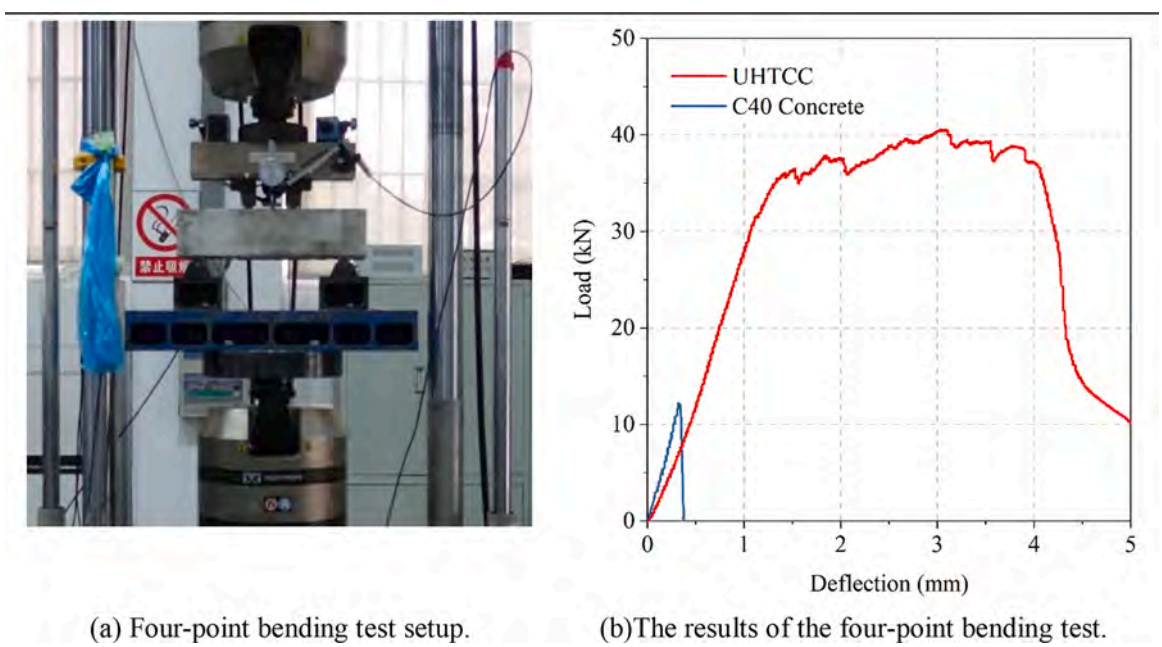


**Fig. 7.** The setup of four-point bending experiments and test results of UHTCC and concrete.

mm. 10,000 Hz samples were employed as the sample rate for LDS. Considering that the fragments from the slabs may destroy the LDS, the LDS was positioned 400 mm from the midpoint of the free boundary of the slabs. Fig. 9 shows the experimental setup.

## 3. Results and discussions

Fig. 10 shows the appearance of the NC-2 and U-2. Cratering damage in the top surface and spalling damage in the bottom surface can be observed. In NC-2, the diameter and depth of cratering were 60.0 cm and 13.5 cm, respectively. And the diameter of spalling was 88 cm. The side surface of the free boundary presented no cracks, which means that just local failure appeared in the NC-2. In U-2, cratering damage in the top surface and swell damage in the bottom surface was observed clearly. The diameter and depth of cratering were 58 cm and 12 cm, respectively. The side surface of the free boundary presented no cracks, and no obvious slab deformation and deflection was observed, which means that only local damage appeared in the U-2. Comparing the U-2 with the NC-2, it is interesting that the pieces of the swell in the bottom surface did not fly out, which implies that the UHTCC behaves an excellent spalling resistance capacity.

Fig. 11 shows the damage failure of NC-4 and U-4. In NC-4, cratering and spalling damage appeared in the top surface and bottom surface, respectively. The diameter and depth of cratering in the top surface were

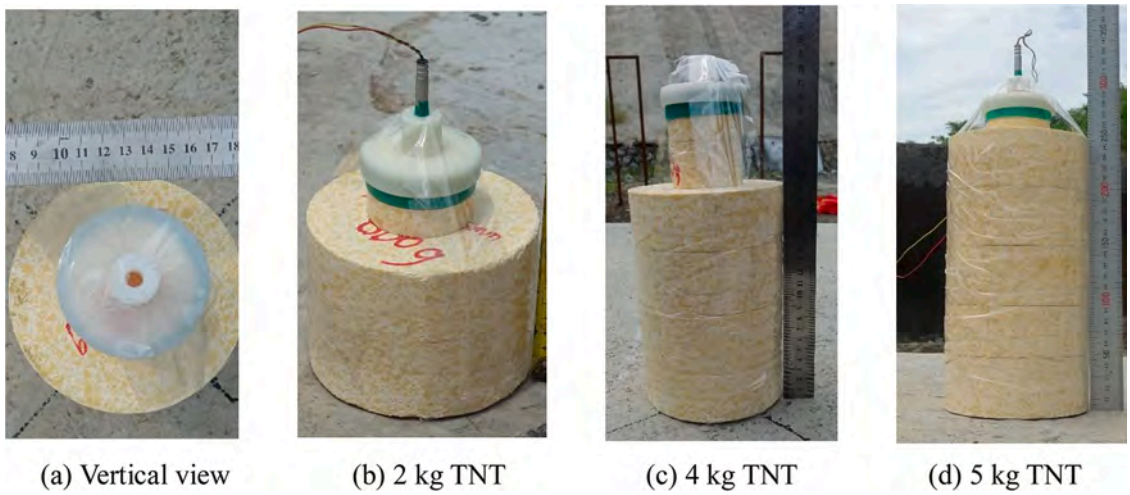


**Fig. 8.** Cylindrical TNT explosives in the tests.

**Table 3**
Test program.

| No. | Specimen IDs | Dimensions (m×m×m) | TNT explosives weight (kg) |
|---|---|---|---|
| 1 | NC-2 | 2.0 × 2.0 × 0.3 | 2 |
| 2 | NC-4 | 2.0 × 2.0 × 0.3 | 4 |
| 3 | NC-5 | 2.0 × 2.0 × 0.4 | 5 |
| 4 | U-2 | 2.0 × 2.0 × 0.3 | 2 |
| 5 | U-4 | 2.0 × 2.0 × 0.3 | 4 |
| 6 | U-5 | 2.0 × 2.0 × 0.4 | 5 |

**NC-2 means that the slab was ordinary concrete (C40), and 2 stands for the weight of TNT (2 kg).

70 cm and 12.7 cm, respectively. And the diameter of spalling in the bottom surface was 95 cm. In addition, cracks appeared in the two side surfaces of NC-4, which means that the NC-4 presented both the local damage and overall response. In addition, no obvious slab deformation and deflection was observed. This may owe the smaller deflection response invisible to observe the slab under 4 kg TNT explosive loading. In U-4, cratering damage appeared on the top surface, and spalling damage appeared on the bottom surface, respectively. The diameter and depth of cratering in the top surface were 50 cm and 11.7 cm, respectively. And the diameter of spalling in the bottom surface was 40 cm. In addition, the side surface of the free boundary presented no cracks, and no obvious slab deformation and deflection was observed in U-4. Comparing U-4 with NC-4, the depth of cratering, the diameter of cratering and spalling of the were much smaller than that of NC-4. That indicates the UHTCC shows better blast resistance performance than ordinary concrete once again.

Fig. 12 shows the damage mode of NC-5 and U-5. In NC-5, cratering and spalling damage appeared in the top surface and bottom surface, respectively. The diameter and depth of cratering in the top surface were 78 cm and 12.5 cm, respectively. And the diameter of spalling in the bottom surface was 118 cm. In addition, cracks appeared in the two side surfaces of the free boundary of NC-5, which means that the local damage and overall response appeared in the concrete slab under 5 kg TNT blast loading. In U-5, cratering damage on the top surface and no

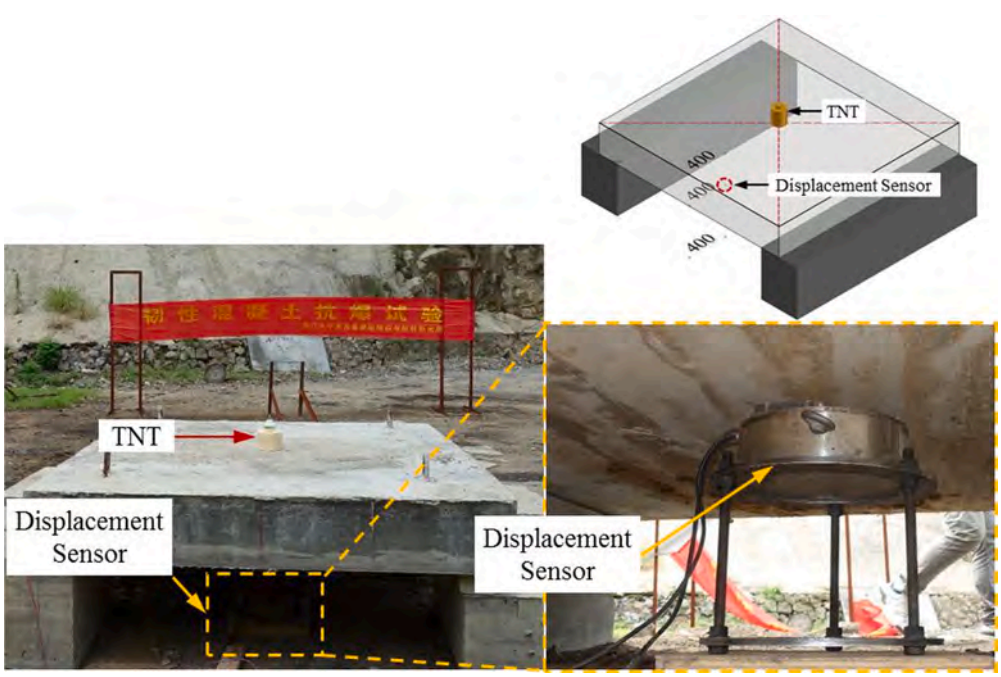


**Fig. 9.** Experimental set-up.

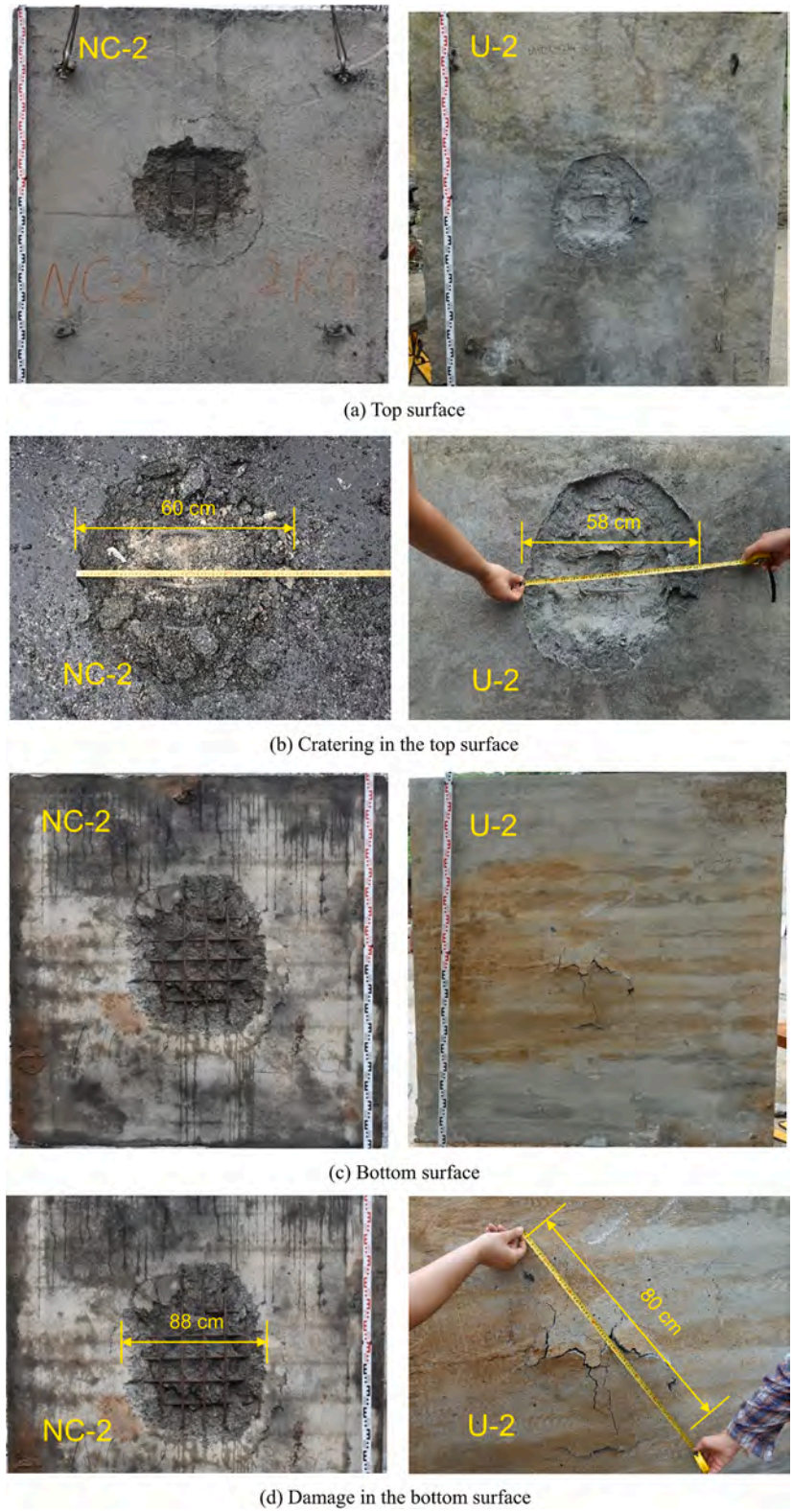


**Fig. 10.** Appearance of NC-2 and U-2.

spalling damage on the bottom surface can be observed. The diameter and depth of cratering were 64 cm and 11 cm, respectively. A few micro-cracks can be observed on the bottom surface. The side surface of the free boundary presented no cracks, and no obvious slab deformation and deflection was observed, which means that only local failure appeared in the U-5.

In contrast, the overall response did not appear. Comparing the U-5 with the NC-5, no spalling damage appeared in the rear surface of the UHTCC slab, while spalling damage appeared in the bottom surface of the ordinary concrete slab. That shows the UHTCC shows better blast resistance performance than ordinary concrete once again.

The damaged area was calculated by Adobe Photoshop through the

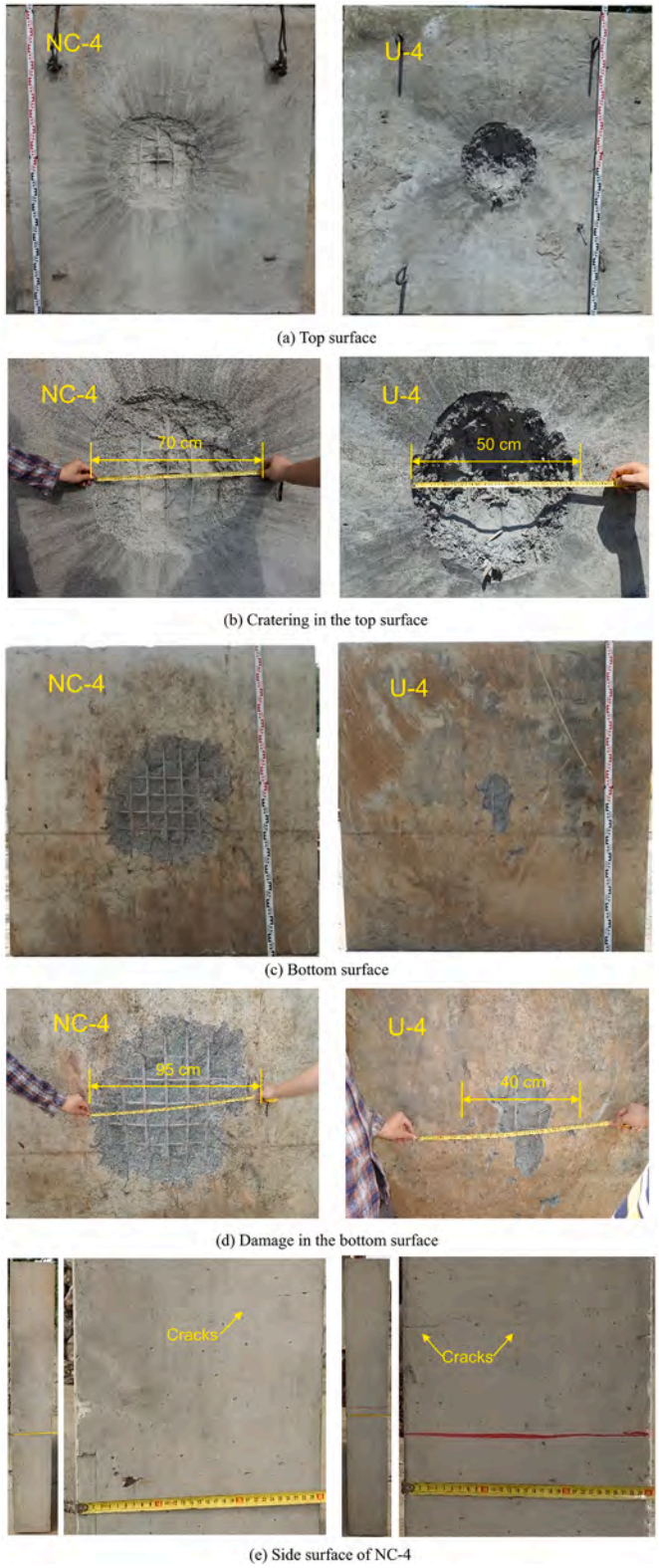


**Fig. 11.** Appearance of NC-4 and U-4.

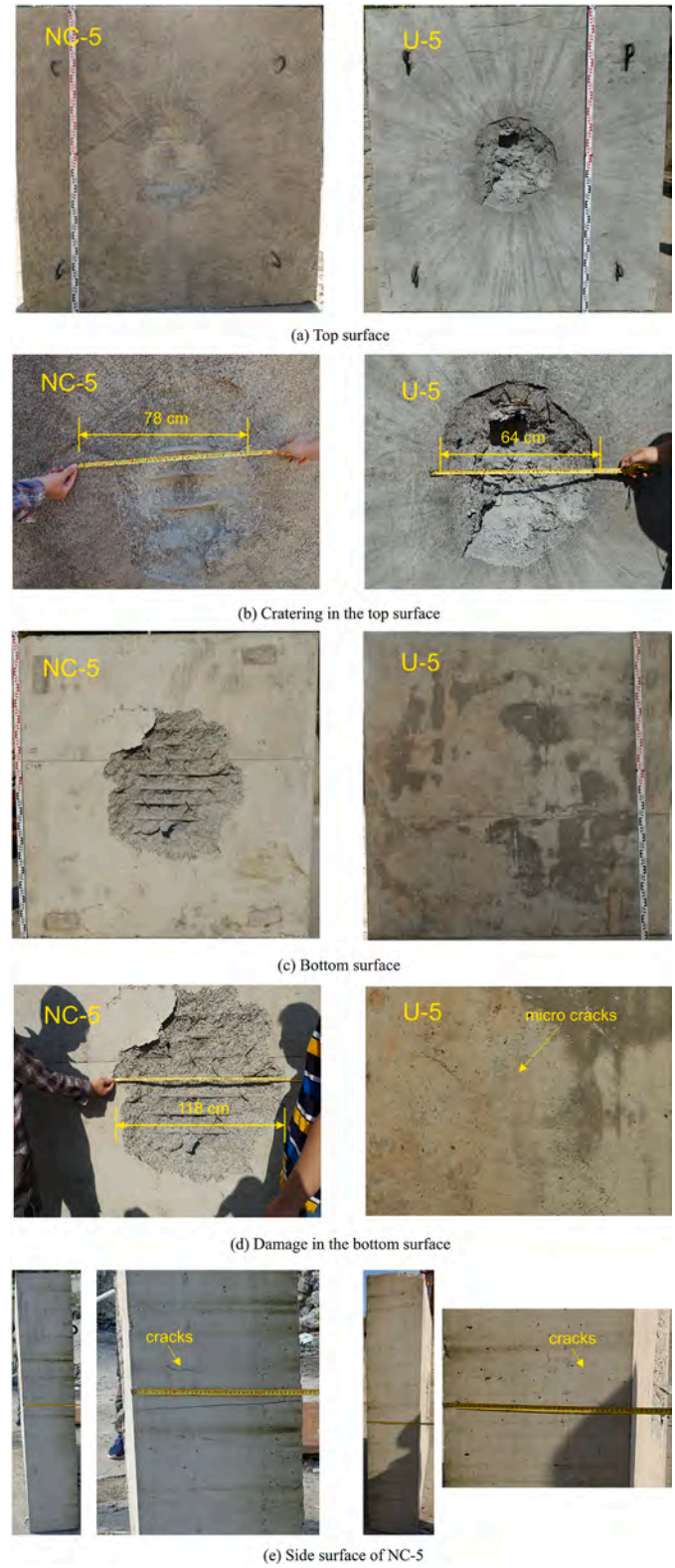


**Fig. 12.** Appearance of NC-5 and U-5.

way of pixel-scale calculating. The results of the damaged area are listed in Table 4. And the damaged area of the top and bottom surface of slabs was depicted in Fig. 13. Comparing the UHTCC and concrete slabs, the cratering area in the top surface and the spalling area in the bottom surface of UHTCC slabs are all smaller than that of concrete slabs. When subjected to 2 kg, 4 kg, and 5 kg TNT blast loadings, the cratering area of concrete slabs is 1.14, 1.67, and 1.18 times as much as UHTCC slabs. And the spalling area of NC-4 is 5.03 times as much as that of U-4. This indicates that the UHTCC could effectively reduce the cratering and spalling damage induced by detonations.

Fig. 14 presents the depth of cratering in the top surface of specimens. The depth of cratering of all UHTCC slabs presented smaller than

**Table 4**
Damaged area of slabs (Unit: $m^2$)

| | 2 kg TNT | | 4 kg TNT | | 5 kg TNT | |
|---|---|---|---|---|---|---|
| | Top | Bottom | Top | Bottom | Top | Bottom |
| NC slab | 0.277 | 0.556 | 0.356 | 0.561 | 0.330 | 0.708 |
| UHTCC slab | 0.244 | 0.000 | 0.213 | 0.062 | 0.279 | 0.000 |

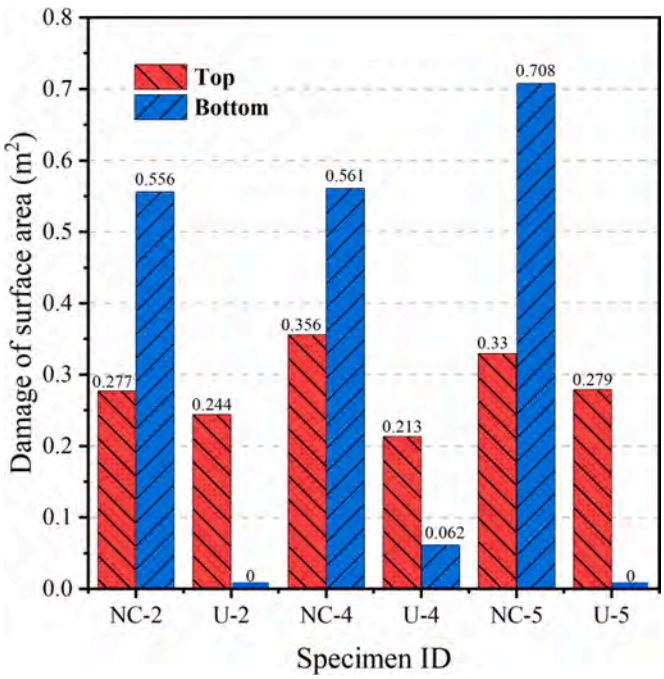


**Fig. 13.** Damaged area of top and bottom surface area of specimens.

that of concrete slabs. In addition, the depth of cratering of concrete and UHTCC slabs decreased with the increase of TNT weight. This phenomenon is consistent with the empirical formulae for predicting the damaged depth in the top surface of slabs. The empirical formula from Yao et al. [55] is as follows.

$$h = K_b W^{1/3} - 0.5R \tag{1}$$

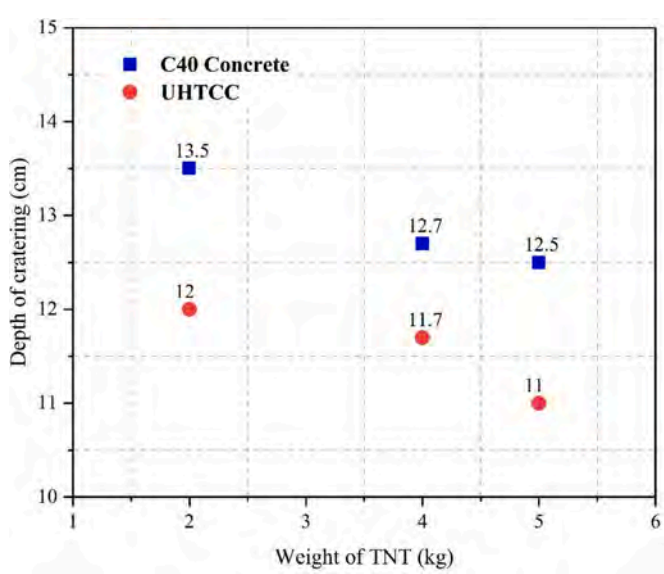


**Fig. 14.** Depth of cratering in the top surface of specimens.

where $K_b$[m/kg$^{1/3}$] is the blast-resistant coefficient, $h$[m] is the cratering depth, $W$[kg] is the weight of TNT, and $R$[m] is the distance from the gravity of charge to the surface of stuctures. In this study, the diameter ($d$) of TNT was 13 cm. Considering the relationship between weight and material density $\rho_c$[kg/m$^3$] of charge, i.e. $W = \rho_c \pi (d/2)^2 H = (\pi/2)\rho_c d^2 R$, $H$[m] is the height of TNT, equation (1) can be written as:

$$h = K_b \left(\frac{\pi}{2}\rho_c d^2\right)^{1/3} R^{1/3} - 0.5R \tag{2}$$

where $K_b\left(\frac{\pi}{2}\rho_c d^2\right)^{1/3}$ is a new constant [cm$^{2/3}$], which can be calculated as 0.347 cm$^{2/3}$ according to Ref. [55]. It should be pointed that the calculated parameter was a reference value according to Ref. [55] just for judging the trend of data change and was not the actual value for UHTCC. More explosive experiments of UHTCC slabs must be conducted to get the experimental data for calculating the blast-resistant coefficient. The function $h = 0.347R^{1/3} - 0.5R$ increases from 0 to 0.1113 and decreases after 0.1113. That means the depth of cratering will decrease with the increase of the height of the center of gravity of the explosive with the same diameter when the height is larger than the critical value.

## 4. Failure mode predictions by empirical equations

To predict the failure mode of UHTCC slabs after blast loadings, one method is to conduct the experiments, and another is to get a better understanding of the mechanical behavior of UHTCC after explosive loadings. The way of predicting contains three methods, namely numerical, theoretical, and empirical prediction. The theoretical prediction depends on some unknown parameters of UHTCC such as the tensile strength of spalling, dynamic fracture energy at varying strain rate and the attenuation coefficient of stress wave propagating in UHTCC, and also depends on some uncertainties, such as the wave dispersion effect, the strain rate effect, and size effect, etc. Therefore, some assumptions and simplifications were employed in the existing research of theoretical predictions [56,57]. However, Li et al. [58] found the theoretical predictions could not accurately predict the failure mode under contact blast loadings.

The numerical prediction depends on the dynamic constitutive model for UHTCC, which is supported by large primary experimental data, such as triaxial tests for strength surfaces and Split Hopkinson Bar (SHPB) tests for compressive and tensile strength under varying strain rates. The numerical simulation will be presented later. And the empirical prediction relies on many tests, and the application of the empirical prediction scopes is limited to situations similar to the data upon which the empirical methods were based. In the present way of empirical predictions, the UFC 3-340-02 report by the US department of defence [59] established the spalling and breaching curves based on the spall and breach data points under blast loadings. In the UFC 3-340-02 report, the spalling curve was plotted by the critical points between the spalling points and non-spalling damage points. Similarly, the breaching curve was plotted by the critical points between the spalling points and the breaching damage points. While the configurations in the tests was different from this study, therefore the prediction curves in UFC 3-340-02 which the specification is applicable to non-contact explosion conditions, and is not available for predicting the failure mode of UHTCC slabs in this study.

McVay [60] put forward an empirical method to predict the damage mode of concrete slabs subjected to blast loadings after analyzing 334 field explosive tests. The scaled distance $R/W^{1/3}$[m/kg$^{1/3}$] and the scald thickness $T/W^{1/3}$[m/kg$^{1/3}$] are the two parameters in McVay's empirical prediction, where $R$[m] is the standoff distance which is the distance from the center of gravity of the explosive to the surface of the concrete slabs, $T$[m] is the thickness of the concrete slabs, and $W$[kg] is the weight of charge. Table 5 lists the corresponding parameters obtained from this study. Note that the parameters of UHTCC slabs were the same

as the concrete slabs in the same experimental condition. Fig. 15 presents the prediction results according to McVay [60]. Combining with the experimental results, McVay [60] can give a good prediction of the damage mode of NC-2, NC-4 and NC-5, where the spalling damage occurred.

U-2 and U-4 presented slightly spalling damage in the blast tests. Bulging and cracking but not flaking appeared in the bottom face of the U-2. And smaller fragments spalling was observed in the bottom surface of U-4. In general, McVay [60] can predict the failure mode of U-2 and U-4. However, only cratering damage in the top surface and no apparent damage in the bottom surface of U-5 can be observed, which shows the same thickness and experimental condition with NC-5. That means the empirical method proposed by McVay [60] underestimates its resistance capacity of spalling and gives an inaccurate prediction. Two factors can explain this observation. One is that McVay's method [60] is based on the data of concrete, which shows different mechanical properties with UHTCC. Another one is that the empirical lines in the graph can only give a fuzzy boundary but not the exact extent of the damage.

Morishita et al. [6] put forward new formulae to predict the damage mode of concrete slabs under contact explosion by combining McVay's method [60] and more test results. The empirical formulae are given as follows:

Cratering: $T/W^{1/3} > 0.36$
Cratering and spall: $0.2 \leq T/W^{1/3} \leq 0.36$
Breaching: $T/W^{1/3} \leq 0.2$

The values and the empirical prediction method of Morishita [6] were plotted in Fig. 16. In the concrete slabs, the behavior of NC-2 and NC-5 is well predicted, and the damage mode of NC-5 is underestimated. While for the UHTCC slabs, only the damage mode of U-2 is predicted but not well. Thus, Morishita [6] underestimate the blast resistance of UHTCC slabs.

McVay's and Morishita's empirical prediction methods underestimate the blast resistance of UHTCC slabs. More influence factors should be paid attention to, such as the reinforcement ratio, the compressive and dynamic tensile strength of materials, and the fracture energy of materials. Furthermore, to more accurately predict the damage severity of UHTCC slabs under blast loadings, more studies, theoretical methods even blast experiments and simulations are needed to modify the prediction method. The test results in this study can be employed to calibrate the numerical model, and the modified model will be used for more simulations.

## 5. Numerical study

### 5.1. Material model of UHTCC

An applicable material model is needed to simulate the response of UHTCC structure and capture the properties of elements of UHTCC structures. A particularly accurate commercial dynamic constitutive model for UHTCC has not been developed currently. The K&C model (MAT72R3) developed and improved by Malvar [61] was employed in this study. This model can be adopted because of two reasons researched by Ranade et.al [62], Xu et.al [63] and Li et.al [64]. The K&C model has three independently fixed "loading surfaces" (yield, maximum, and residual). Another one is that this dynamical model has independent parameters for controlling the damage evolutions in tension and compression. The K&C model employed in the numerical simulation study is introduced briefly in the following sections.

**Table 5**
Parameters for failure mode predictions

| Specimen IDs | $T$ (m) | $W$ (kg) | $R$(m) | $T/W^{1/3}$(m/kg$^{1/3}$) | $R/W^{1/3}$(m/kg$^{1/3}$) | Damage mode |
|---|---|---|---|---|---|---|
| NC-2 | 0.3 | 2 | 0.0551 | 0.238 | 0.0437 | Spalling |
| NC-4 | 0.3 | 4 | 0.1111 | 0.189 | 0.0700 | Spalling |
| NC-5 | 0.4 | 5 | 0.1293 | 0.234 | 0.0765 | Spalling |
| U-2 | 0.3 | 2 | 0.0551 | 0.238 | 0.0437 | Spalling |
| U-4 | 0.3 | 4 | 0.1111 | 0.189 | 0.0700 | Spalling |
| U-5 | 0.4 | 5 | 0.1293 | 0.234 | 0.0765 | No Spalling |

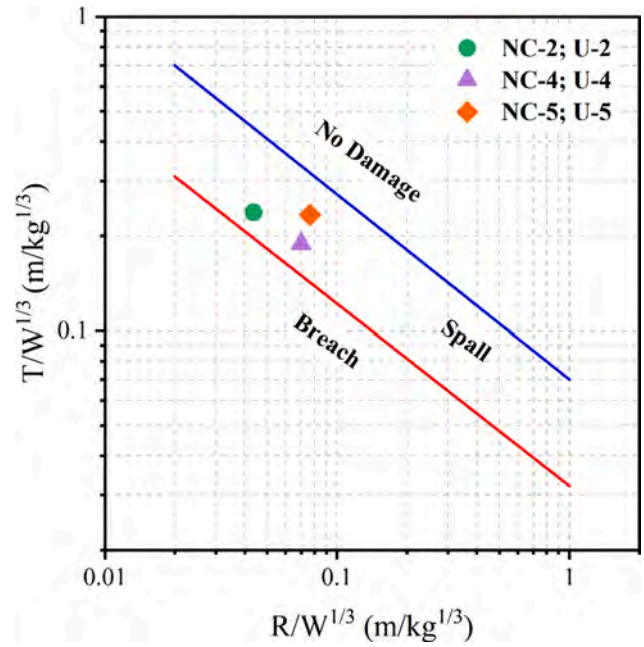


**Fig. 15.** Empirical prediction by McVay [60].

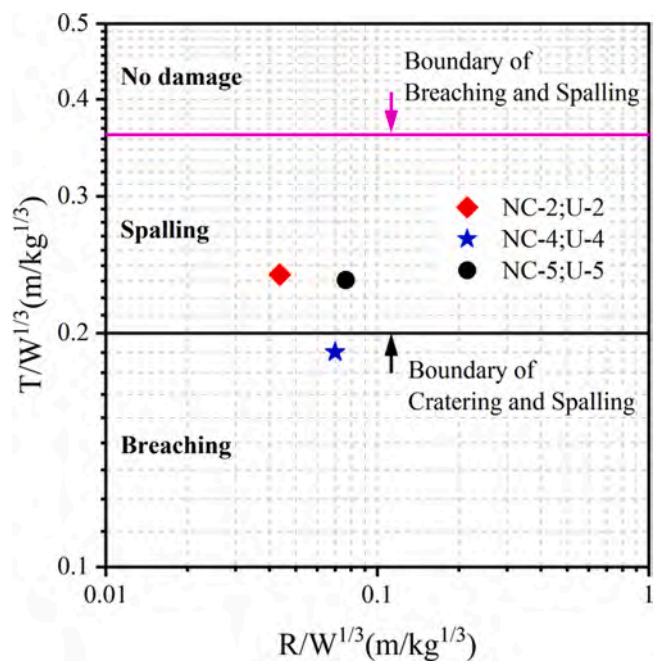


**Fig. 16.** Empirical prediction by Morishita et al. [6].

#### 5.1.1. The governing equations and strength surfaces

The response of concrete structures subjected to blast loadings is a large deformation. When solving this problem, the updated Lagrangian equation is employed to describe the weak form of the equation of motion of concrete structures as follows.

$$\int_{\Omega_x} \rho_u \delta\mathbf{u}\cdot\ddot{\mathbf{u}} d\Omega + \int_{\Omega_x} \left(\nabla_x^s \delta\mathbf{u}\right)\cdot\boldsymbol{\sigma} d\Omega - \int_{\Omega_x} \delta\mathbf{u}\cdot\mathbf{F} d\Omega - \int_{\partial\Omega_x^t} \delta\mathbf{u}\cdot\mathbf{t} d\Omega = 0 \tag{3}$$

where $\rho_u$[kg/m$^3$] is the mass density of UHTCC; $\mathbf{u}$ is the displacement vector; overhead dot = time differentiation; $\boldsymbol{\sigma}$ is the Cauchy stress tensor; $\nabla_x^s$ is the symmetric spatial gradient operator; $\mathbf{b}$ is the body force; $\mathbf{t}$ is the traction on the natural boundary $\partial\Omega_x^t$; $\Omega_x$ is the problem domain. And all the variables are evaluated in the deformed configuration. The constitutive laws employed to simulate the response of concrete mainly govern the evolution of the stress state $\boldsymbol{\sigma}$. In the K&C model, the failure behavior is characterized by a pressure-dependent yield-surface $F(p,\boldsymbol{\sigma},\lambda)$ [61], which is written as follows.

$$F(p,\boldsymbol{\sigma},\lambda) = \sqrt{3J_2} - \Gamma(p, J_3, \lambda) \tag{4}$$

where $J_2$ and $J_3$ are the second and third invariant of the deviatoric stress tensor $\boldsymbol{\sigma}'$, respectively; $J_2 = \frac{1}{2}\boldsymbol{\sigma}'\cdot\boldsymbol{\sigma}'$ and $\boldsymbol{\sigma}' = \boldsymbol{\sigma} + p\mathrm{I}$. $p$ is the hydrostatic pressure and equal to $-\boldsymbol{\sigma}_{kk}/3$. $\lambda$ is the damage parameter to describe the damage effect on the behavior of materials.

The function $F(p,\boldsymbol{\sigma},\lambda)$ is the failure strength surface in current state ($p$, $\lambda$). In the K&C model, the current failure strength surface is determined by interpolating through three strength surfaces, i.e. the maximum strength surface, yield strength surface and residual strength surface. The interpolating function is as follows.

$$F(p,\sigma,\lambda) = \begin{cases} r_f\cdot\Theta(J_3)\cdot\left[\eta(\lambda)\cdot\left(\Delta\sigma_m - \Delta\sigma_y\right) + \Delta\sigma_y\right] & \lambda \le \lambda_m \\ r_f\cdot\Theta(J_3)\cdot\left[\eta(\lambda)\cdot\left(\Delta\sigma_m - \Delta\sigma_r\right) + \Delta\sigma_r\right] & \lambda \ge \lambda_m \end{cases} \tag{5}$$

where $r_f$ [-] is the dynamic increase factor. $\eta(\lambda)$ is the damage evolution function which changes from 0.0 to 1.0. $\lambda_m$[-] is corresponding to $\eta$=1.0. $\Delta\sigma_y$ is the yield strength surface. $\Delta\sigma_m$ is the maximum strength surface. $\Delta\sigma_r$ is the residual strength surface. Eq. (5) is also shows the computing method in the load and unload process. The three strength surfaces are as follows.

$$\begin{cases} \Delta\sigma_y = a_{0y} + p/\left(a_{1y} + a_{2y}p\right) \\ \Delta\sigma_m = a_0 + p/(a_1 + a_2 p) \\ \Delta\sigma_r = a_{0f} + p/\left(a_{1f} + a_{2f}p\right) \end{cases} \tag{6}$$

where $a_0$[MPa], $a_1$[-], $a_2$[MPa$^{-1}$], $a_{0y}$[MPa], $a_{1y}$[-], $a_{2y}$[MPa$^{-1}$], $a_{0f}$[MPa], $a_{1f}$[-] and $a_{2f}$[MPa$^{-1}$] are the parameters of strength surfaces. The parameters were determined according to the triaxial experimental data in the previous study [63]. $p$[MPa] is the hydrostatic pressure. $\Delta\sigma = \sqrt{3J_2}$, and $J_2$ is the second invariant of the deviatoric stress tensor.

Fig. 17 shows three strength surfaces, as well as the load and unload process in K&C model. Fig. 17-(a) presents the load and unload path. Fig. 17-(b) present three typical points corresponding to the points in Fig. 17-(a). In the load process, the strength is obtained by interpolation calculation from the yield surface and maximum surface. In the unload process, the strength surface is obtained by interpolation calculation from the maximum surface and the residual surface, as calculated in Eq. (5). In Eq. (5), $\eta$ is the yield scale factor related to the damage function $\lambda$, which is determined by the following formulas:

$$\lambda = \begin{cases} \int_0^{\overline{\varepsilon^p}} \frac{\overline{d\varepsilon^p}}{r_f\left(1 + p/r_f f_t\right)^{b_1}} \text{for} p \ge 0 \\ \int_0^{\overline{\varepsilon^p}} \frac{\overline{d\varepsilon^p}}{r_f\left(1 + p/r_f f_t\right)^{b_2}} \text{for} p \le 0 \end{cases} \tag{7}$$

where $\overline{d\varepsilon^p} = \sqrt{\frac{2}{3}d\varepsilon_{ij}^p d\varepsilon_{ij}^p}$ is the effective plastic strain increment. $b_1$[-] and $b_2$[-] are the damage scaling parameters controlling the rates of damage accumulation $\eta$[-] for compression and tension respectively. According to the previous study [63], And the value of $b_1$ and $b_2$ are set equal to 2.4 and -11.3, respectively. The previous study [63] gives the $\eta$-$\lambda$ damage function is input by the user given in Table 6 based on the material's experimental uniaxial compressive behavior.

### 5.1.2. Equation of state (EOS)

An equation of state (EOS) for UHTCC is also employed to describe the relationship between the hydrostatic pressure and the volumetric strain. In this study, the EOS-8 in LS-DYNA is employed.

$$p = C(\mu) + \gamma_0\theta(\mu)E_0 \tag{8}$$

where $C(\mu)$[MPa] and $\theta(\mu)$[-] are the tabulated pressure evaluated along a 0 K isotherm and tabulated temperature-related parameter as functions of the volumetric strain, respectively. According to [62,63,64,65], the temperature-dependent variation of hydrostatic pressure was neglected. Therefore, the EOS 8 used for UHTCC is automatically generated for the similar compressive strength concrete, which assumes that UHTCC has the same properties as concrete under hydrostatic pressure. Fig. 18 illustrates the load and unload process in EOS 8. The hydrostatic pressure is determined by calculating through Eq. (11) in the load process. While in the unload process, it occurs along the unload path with the unload bulk modulus to the pressure cutoff. And reload

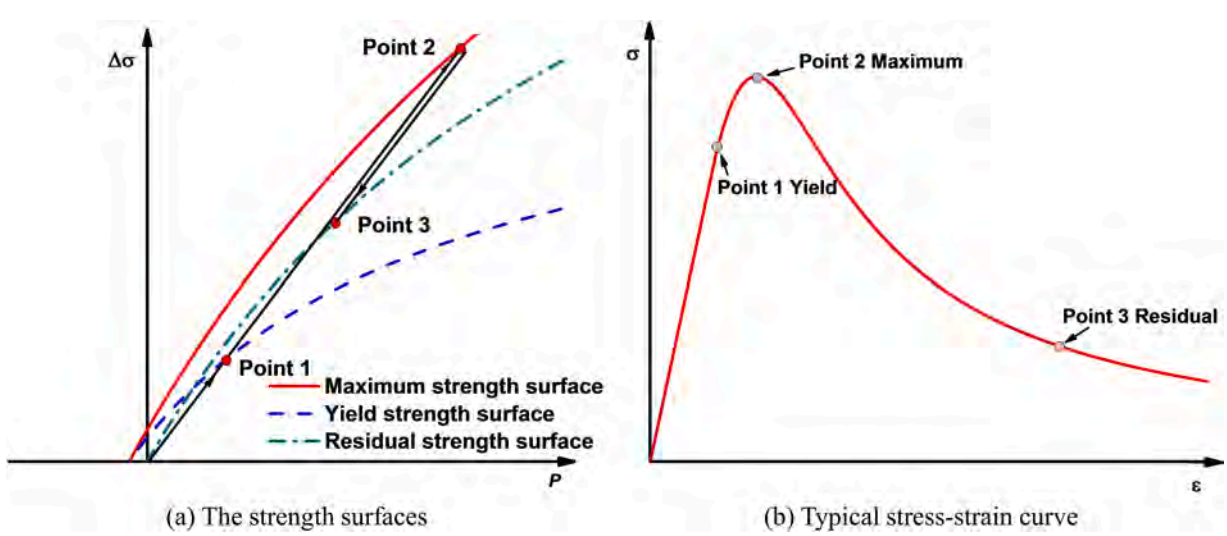


**Fig. 17.** Strength model for UHTCC.

**Table 6**
The defining points for the $\eta$-$\lambda$ damage function.

| $\lambda$ | 0.0 | $8 \times 10^{-6}$ | $2.4 \times 10^{-5}$ | $4 \times 10^{-5}$ | $5.6 \times 10^{-5}$ | $7.2 \times 10^{-5}$ | $8.8 \times 10^{-5}$ | $3.2 \times 10^{-4}$ | $5.2 \times 10^{-4}$ | $5.7 \times 10^{-4}$ | 1.0 | 10 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\eta$ | 0.0 | 0.85 | 0.97 | 0.99 | 1.0 | 0.99 | 0.97 | 0.5 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 |

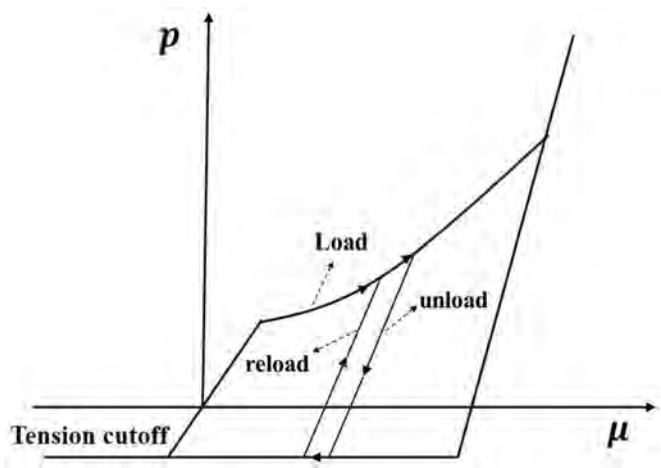


**Fig. 18.** Schematic diagram of the EOS-8.

occurs along the unload path to the state where the unloading begins. The previous study [63] presents the parameters generated by EOS-8 based on the elastic modulus and Poisson's ratio of UHTCC as listed in Table 7.

### 5.1.3. Rate effects

The strain rate effects have a noticeable effect on the dynamical behavior of UHTCC [66]. Generally, the dynamic increase factor is employed to describe the amplification of strength by impact, which is equal to the dynamic strength divided by the static strength. Researchers [67,68,69] found that the strain rate effect on the tensile performance of UHTCC is different from ordinary concrete, and strain-hardening could still be observed in the dynamic tensile tests. And previous study [63] consider the compressive and tensile strain rate based on the study conducted by Gebbeken and Ruppert [2], Tu and Lu [70], and Xu and Wen [71].

$$r_{ft} = \left\{[\tanh(\log(\dot{\varepsilon}_u / \dot{\varepsilon}_0) - W_{xt})S_t]\left(F_{mt} / W_{yt} - 1\right) + 1\right\}W_{yt} \tag{9}$$

$$r_{fc} = \left\{[\tanh(\log(\dot{\varepsilon}_u / \dot{\varepsilon}_0) - W_{xc})S_c]\left(F_{mc} / W_{yc} - 1\right) + 1\right\}W_{yc} \tag{10}$$

where $F_{mt}$, $W_{xt}$, $S_t$, $W_{yt}$, $F_{mc}$, $W_{xc}$, $S_c$, $W_{yc}$ are the parameters in formulas (9) and (10). All the eight parameters are dimensionless and can be obtained by fitting the existing experimental data. The values of those parameters are listed in Table 8 in Ref. [63].

In the K&C model, the strength surfaces and the damage accumulation function were enhanced by the strain rate effect as follows:

$$\Delta\sigma = r_f \Delta\sigma(p / r_f) \tag{11}$$

$$\lambda = \lambda(p / r_f) / r_f \tag{12}$$

**Table 8**
Parameters of dynamic increase factor formulae of UHTCC.

| $F_{mt}$ | $W_{xt}$ | $S_t$ | $W_{yt}$ | $F_{mc}$ | $W_{xc}$ | $S_c$ | $W_{yc}$ |
|---|---|---|---|---|---|---|---|
| 10 | 1.8 | 0.85 | 5.5 | 2.94 | 2.45 | 1.52 | 1.97 |

While the strain rate effect on UHTCC's tensile behavior is complex, such as on the softening stage and the deformation value of strain hardening. Some researches [24,67–69,72–76] attempted to solve the relative problems, but a relative unified conclusion still cannot be obtained. Therefore, the strain rate effect on the dynamic tensile state of UHTCC cannot be explicitly specified currently in LS-DYNA, and the numerical simulation in the current study did not consider this factor. In addition, the erosion criterion is chosen to simulate the UHTCC's properties at the highest strain rate and hydrostatic pressure. Ranade et al. [62] suggested the erosion criterion for UHTCC, where the tensile strain was set at 0.04 for maximum tensile strain and the hydrostatic pressure was set at 400 MPa.

## 5.2. Material model for reinforcement steel

The *MAT_PLASTIC_KINEMATIC (MAT_03) in LS-DYNA was employed for the reinforcement steel, which is suitable for modelling the isotropic and kinematic hardening plasticity with the option of including the rate effects. The model is described as follows:

$$\sigma_{ys} = \left[1 + \left(\frac{\dot{\varepsilon}}{C}\right)^{\frac{1}{p}}\right]\left(\sigma_0 + \beta E_p \varepsilon_{eff}^p\right) \tag{13}$$

where $\sigma_{ys}$[MPa] is the yield strength, $\sigma_0$[MPa] is the initial yield strength. $E_p$[GPa] is the plastic hardening modulus. $\varepsilon_{eff}^p$ is the effective plastic strain and $\beta$[-] is the relative parameter. $p$[-] and $C$[-] are the parameters related to strain rate and are set as 5.0 and 40.0, respectively, as in Ref. [65].

## 5.3. Material model for air and TNT

In this study, a simple equation (EOS-1) of the state of the ideal gas law was employed for air in LS-DYNA. The pressure is related to the specific internal energy.

$$p = (\gamma - 1)\rho_a e_a \tag{14}$$

where $\gamma$[-], $\rho_a$[kg/m$^3$] and $e_a$[kJ/kg] is the constant, density of air and the specific internal energy, respectively. The standard parameter ($\gamma$ = 14, $\rho_a$= 1.225 kg/m$^3$) of air was employed in the study. And the air initial internal energy is assumed to be 2.068 × 10$^5$ kJ/kg, as in Ref. [65].

In addition, the JWL equation was employed for the TNT. The pressure generated by the chemical energy of TNT can be calculated

**Table 7**
Parameters generated automatically in EOS-8.

| $\mu_1$ | $\mu_2$ | $\mu_3$ | $\mu_4$ | $\mu_5$ | $\mu_6$ | $\mu_7$ | $\mu_8$ | $\mu_9$ | $\mu_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.0015 | 0.0043 | 0.0101 | 0.0305 | 0.0513 | 0.0726 | 0.0943 | 0.174 | 0.208 |
| $K_{av1}$ | $K_{av2}$ | $K_{av3}$ | $K_{av4}$ | $K_{av5}$ | $K_{av6}$ | $K_{av7}$ | $K_{av8}$ | $K_{av9}$ | $K_{av10}$ |
| $K_v$ | $K_v$ | 1.014 $K_v$ | 1.065 $K_v$ | 1.267 $K_v$ | 1.470 $K_v$ | 1.672 $K_v$ | 1.825 $K_v$ | 4.106 $K_v$ | 5.000 $K_v$ |
| $p_1$ | $p_2$ | $p_3$ | $p_4$ | $p_5$ | $p_6$ | $p_7$ | $p_8$ | $p_9$ | $p_{10}$ |
| 0 | $K_v \times \mu_2$ | 2.18 $p_2$ | 3.5 $p_2$ | 6.65 $p_2$ | 10.03 $p_2$ | 14.23 $p_2$ | 21.77 $p_2$ | 127.1 $P_2$ | $p_2$ |

from the JWL equation, which is written as follows.

$$p = A\left(1 - \frac{\omega}{R_1 V}\right)e^{-R_1 V} + B\left(1 - \frac{\omega}{R_2 V}\right)e^{-R_2 V} + \frac{\omega E}{V} \tag{15}$$

where $p$[MPa] is the hydrostatic pressure. $A$[MPa], $B$[MPa], $R_1$[-], $R_2$[-] and $\omega$[-] are the TNT parameters. $V$[m$^3$] and $E$[kJ/kg] are the specific volumes and internal energy, respectively. In the simulation, $A$, $B$, $R_1$, $R_2$ and $\omega$ are set at $3.7377 \times 10^5$ MPa, $3.747 \times 10^3$ MPa, 4.15, 0.9, and 0.35 respectively, as same as in Ref. [77].

### 5.4. Finite element (FE) model

In the numerical simulation, the material parts can be distinguished as UHTCC, air, TNT, steel and support. Different parts consisted of different types of elements. The Euler solid 3D elements with 20 mm mesh size were used to simulate the air and TNT. The TNT was embedded into the air by using the card *INITIAL_VOLUME_FRACTION_GEOMETRY. The part *CONSTRAINED_LAGRANGE_IN_SOLID was used to solve the interaction of UHTCC with air and TNT by using the part. The steel bars were meshed with Hughes-liu beam [78] elements embedded in UHTCC. This model is suitable for modelling the isotropic and kinematic hardening plasticity with rate effects. No bond-slip relationship was assumed between the steel reinforcements and the UHTCC due to the short time of blast [79]. The *CONSTRAINED_LAGRANGE_IN_SOLID card was used to make the UHTCC slab and the steel rebar together. This command provide no bond-slip between the UHTCC part and steel part. The UHTCC were meshed with 3D Lagrange solid elements. For the FE model of UHTCC, two kinds of mesh refinement were established, including the structured and unstructured discretizations. It is well known that the mesh size can affect the precision of FE results. To investigate the influence of the mesh size, three types of mesh size were employed, i.e. 10 mm, 20 mm and 50 mm. Fig. 19 shows the structured and unstructured discretizations with mesh size of 50 mm. The boundary condition of the Euler subgrid is set as an outflow boundary by usng the command *BOUNDARY_NON_REFLECTING. The supports were meshed with Lagrange solid elements, which was described by using the K&C model of ordinary concrete. And the contact between the support and the UHTCC slab were linked by using the command *CONTACT_AUTOMATIC_SURFACE_TO_SURFACE. The erosion technique was employed to simulate the serious damage process, such as breaching and spalling in the bottom surfaces. The erosion criterion where the tensile strain was set at 0.04 for maximum failure strain and the hydrostatic pressure with 400 MPa were suggested based on the research [62]. In addition, the influence of the maximum falilre strain on the results will be investigated in section 6. Fig. 20 presents the FE model.

### 5.5. Numerical results

Figs. 21–23 present the effective plastic strain obtained from the unstructured and structured discretizations in the top surface of UHTCC slabs under 2 kg, 4 kg, and 5 kg TNT contact explosions. All the damage remains stable and implies that the slab response under contact explosion is highly localized. In the experimental observation, cratering damage in the top surface of all slabs can be observed. And numerical model simulated the upper cratering of all UHTCC slabs. For the structured and unstructured discretizations, the eroded areas in the top surface of UHTCC slabs decrease with the increase of mesh size. This is because the failure energy of elements increases with the increase of

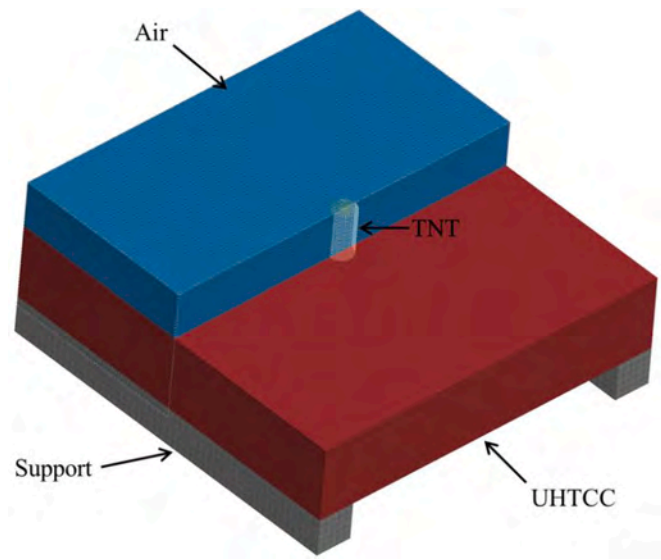


**Fig. 20.** FE model.

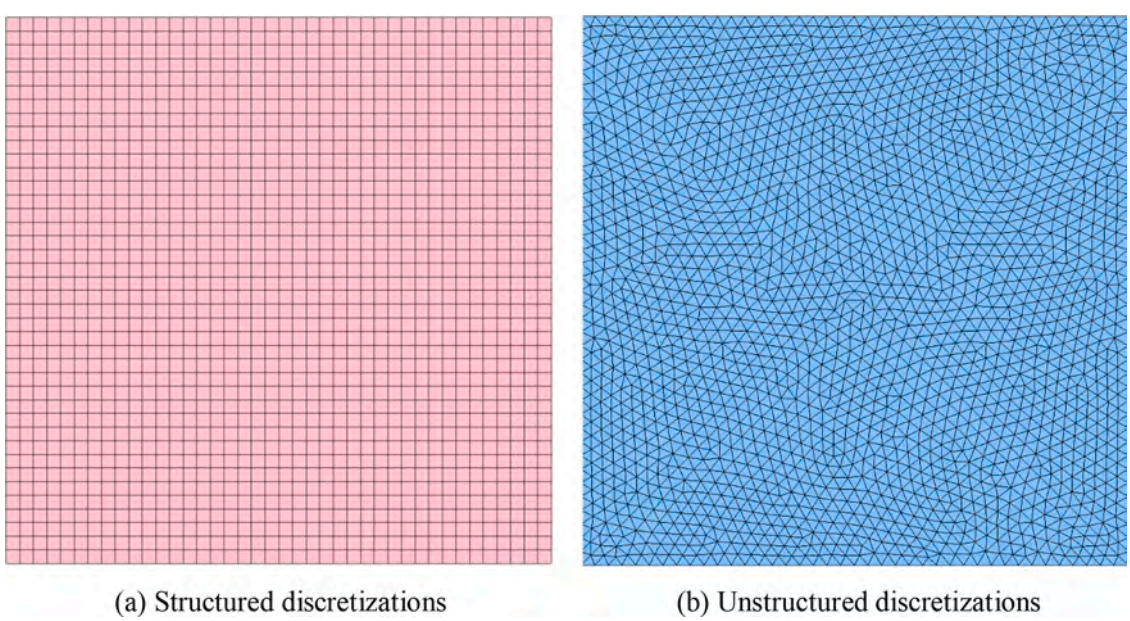


**Fig. 19.** Structured and unstructured discretizations with mesh size of 50 mm.

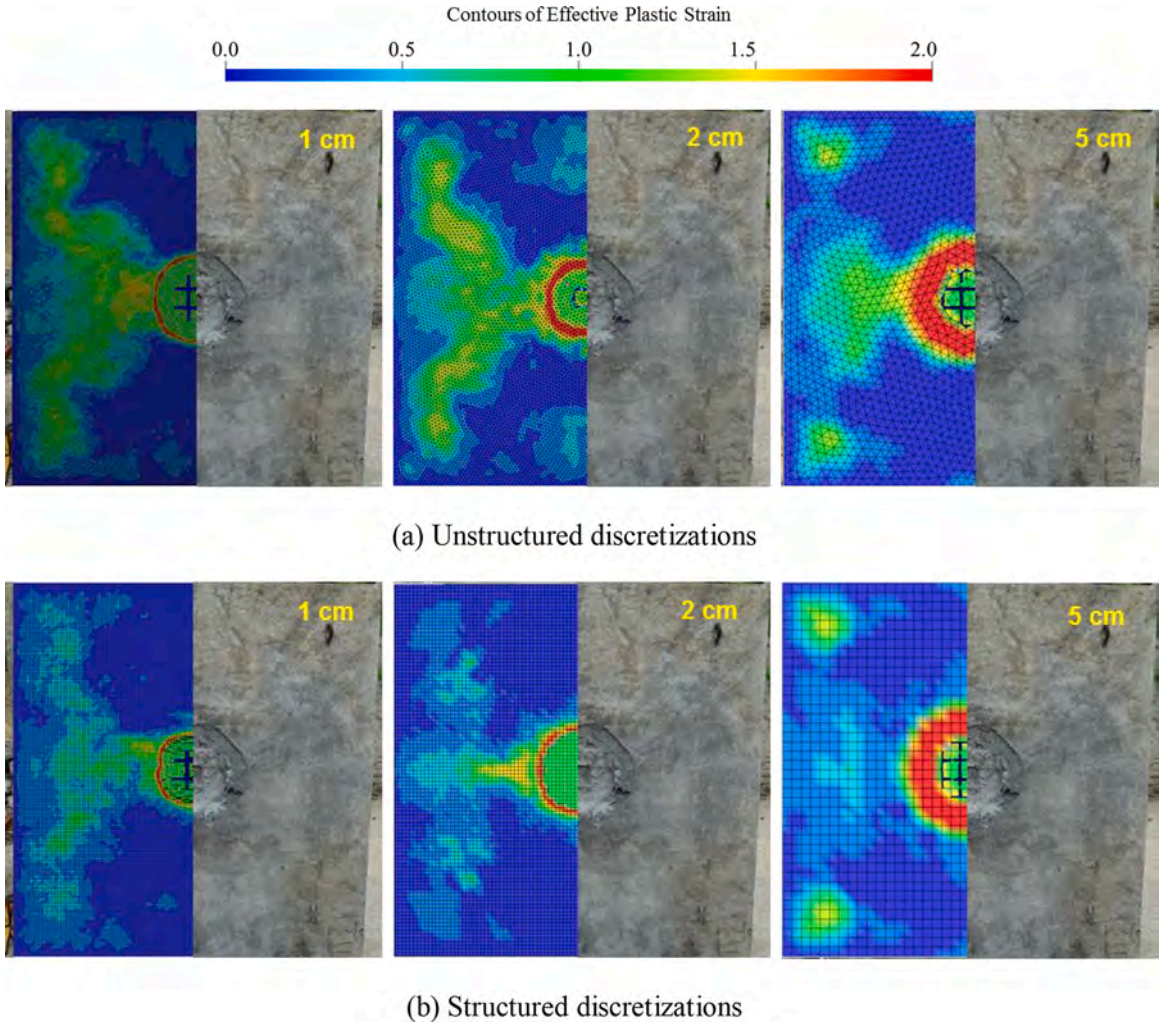


**Fig. 21.** Comparison results from simulations and experiments of U-2.

element size. In addition, the shape of cratering in the top surface from the unstructured discretizations is close to circular while that from the structured discretizations is close to square. That means the unstructured discretization method is more suitable for simulation the dynamic response of UHTC slabs subjected to blast loadings.

Table 9 presents the comparisons between the numerical and experimental results. Fig. 24 displays the damaged area and depth errors of results from unstructured and structured discretizations. When the mesh size is 10 mm, the damaged area error from the unstructured discretizations are all smaller than 20%, while that from the structured discretizations are all larger than 20%. And the damaged depth from the unstructured discretizations are all smaller than or equal to 25%. That means the unstructured discretizations are more suitable for simulating the dynamic behavior of UHTCC slabs when subjected to blast loadings. When the mesh size is 20 mm, the damaged area errors of results from the unstructured discretizations are significantly larger than that from the structured discretizations. However, the damaged depth errors from the unstructured discretizations are smaller than that from the structured discretizations. When the mesh size is 50 mm, the error of results from the structured and unstructured discretizations shows closely or equally in damaged area and depth. This is because the damaged depth from the numerical results mainly depends on the mesh size and it presents multiple relationship with the mesh size.

Although the error of numerical results from the unstructured discretizations with 10 mm mesh size show better prediction than the results from other set up, the prediction errors are still larger than 10%. Some factors could be employed to explain this phenomenon. From the constitutive model of materials view, the EOS used in this study is also auto-generated based on ordinary concrete, which can not relflect the behavior of UHTCC. In the future work, experiments should be conducted to investigate the EOS of UHTCC. And many attempts should also be made to modify the constitutive model, such as the strain rate effect on the strain hardening of UHTCC and the softening behavior in the tension of the UHTCC. From the FE model of view, the input parameters of the model such as the erosion criterion could affect the numerical results directly. All the described uncertainties could affect the predicted results.

Fig. 25 presents the deflection-time curves of slabs from structured and unstructured discretizations with different mesh size. It is regretted the displacement-time curves could not be acquired because the LDS was broken. Therefore, the results from the structured and unstructured discretizations are compared and analyszed in this section. All the peak deflections of slabs increase with the increase of mesh size, and it is because the motion of element increases with the element size when in the same load. The peak deflection from the unstructured discretizations was significantly less than that from the structured discretizations in the same numerical condition, and it is because the volume of the unstructured element is smaller than that of the structured for the same mesh size.

Fig. 26-(a)-(c)shows the eroded volume fraction evolution with time in the simulation for slabs from the structured and unstructured discretizations with different mesh size.

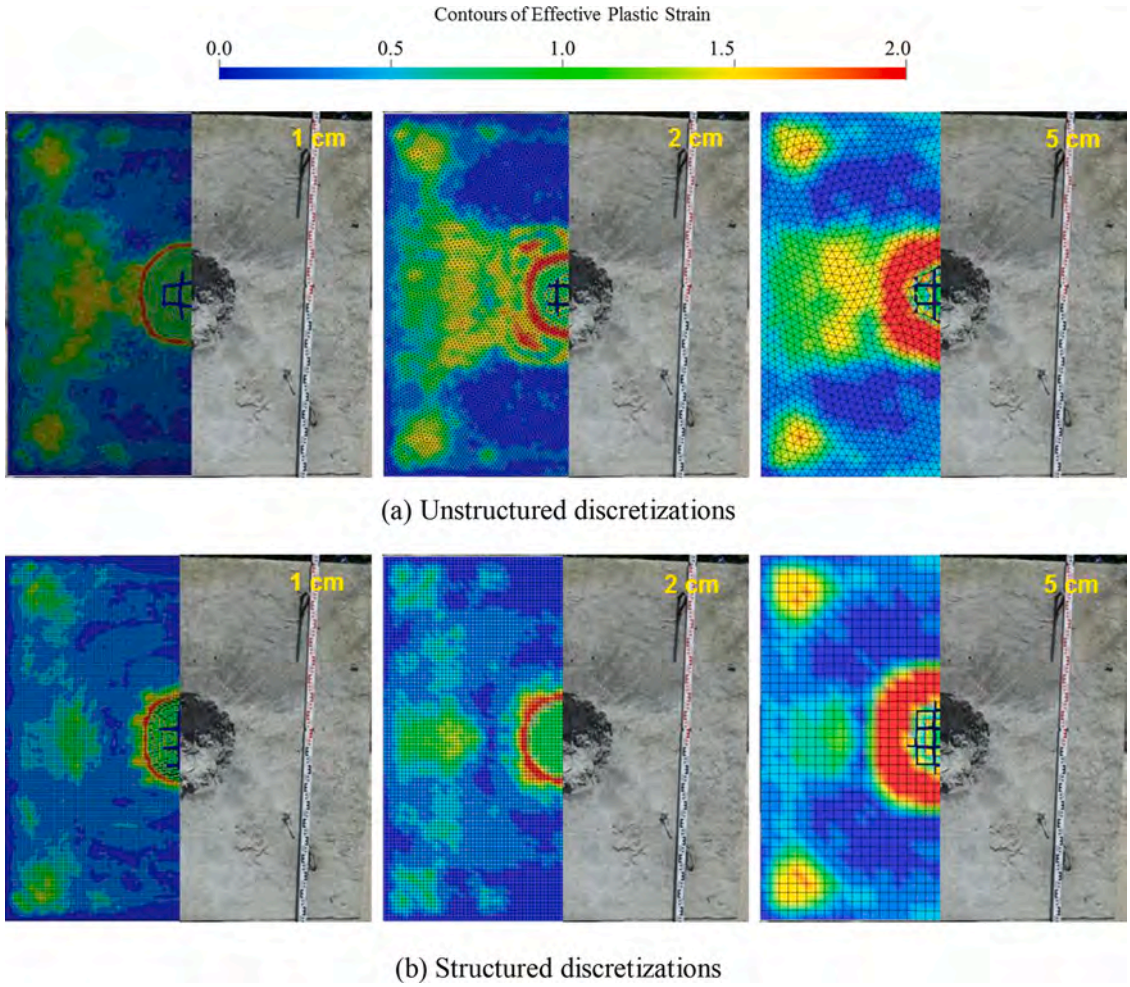


**Fig. 22.** Comparison results from simulations and experiments of U-4.

Fig. 26-(d) is the histogram of eroded volume of different cases. For the unstructured discretizations, the eroded volume fraction with 10 mm mesh size was larger than that with 20 mm and 50 mm in the termination of simulatons. And the eroded volume fraction with 20 mm mesh size was smaller than that with 10 mm and 50 mm. Whereas, the results of the eroded volume fraction from the structured discretizations presented unconspicuous patterns. When the mesh size is 10 mm, the eroded volume fractions from unstructured discretizations are larger than that from structured discretizations. When the mesh size is 50 mm, the eroded volume fractions from structured discretizations are larger than that from unstructured discretizations.

## 6. Uncertainty analysis

When conducting the numerical simulations, various uncertain parameters could affect the numerical results. The uncertain parameters includes two types, i.e. the mechanical model and the material properties. Therefore, a comprehensive study is essential to measure the influence of the uncertainties in the input parameters. Usually, the uncertainty analysis can be achieved by means of sensitivity analysis (SA), including the local SA and global SA. The global SA aims at quantifying the significance of the input parameters. PCE-Sobol method and the PAWN method are the common approaches for global SA [80]. Hamdia et al. [50] employed the PCE model to conduct stochastic analysis of the fracture toughness of polymeric nanoparticle composites, and developed second and third order PCE. In this section, the PCE surrogate model is also employed to study the uncertainties of the input parameters in the numerical simulations.

### 6.1. Polynomial chaos expansions (PCE)

Considering the computational model having $n$ input independent parameters, the output value $y$ is a function of the input parameters, i.e. $y = f(\mathbf{X})$, where $\mathbf{X} = \{X_1,\ldots,X_n\}$. According to [51,81], any second-order random variable may be expanded by a spectral expansion using orthogonal polynomials as follows:

$$Y = \sum_{j=0}^{\infty} Z_j \Psi_j(\xi) \tag{16}$$

where $\boldsymbol{\xi}$ is the vector of independent standard normal (Gaussian) random, and $\Psi_j$ is the orthogonal with respect to the Gaussian measure. $Z_j$ are the PCE deterministic coefficients. When the input random variables are standard normal distributons $\mathbf{X} = \boldsymbol{\xi}$, a possible Hilbertian basis is the family of multivariate Hermite polynomials, which are orthogonal with respect to the Gaussian measure. By employing the usual notation for the multivariate Hermite polynomials, Eq. (16) can be written as:

$$Y = \sum_{j=0}^{\infty} Z_j \Psi_j(\xi_1, \ldots, \xi_n) \tag{17}$$

For computation, Eq. (17) shall be truncated to a polynomial of order $N$. Considering all N-dimensional Hermite polynomials of degree not exceeding $P$, Eq. (17) could be written as follows:

$$Y = \sum_{j=0}^{P-1} Z_j \Psi_j(\xi_1, \ldots, \xi_N) \tag{18}$$

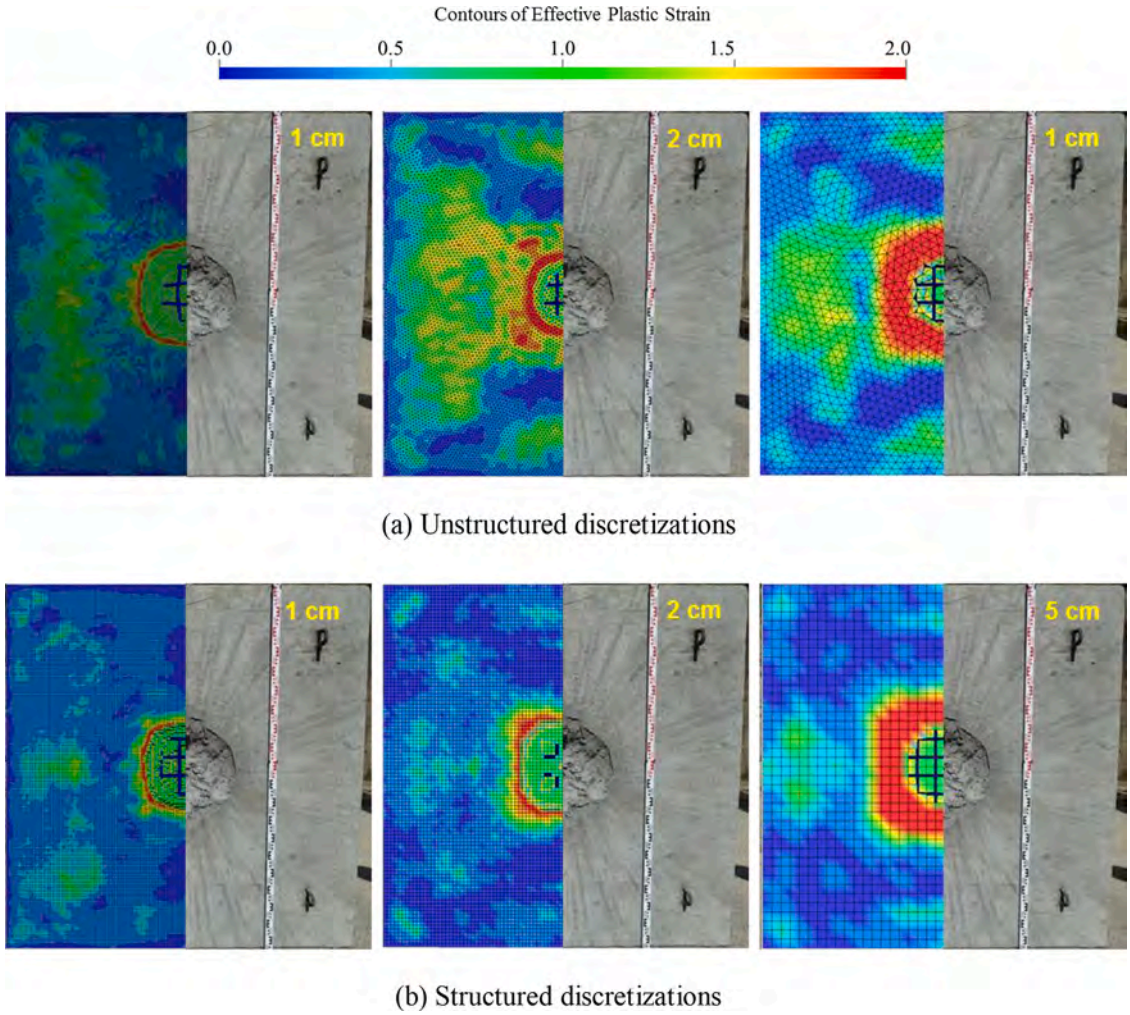


**Fig. 23.** Comparison results from simulations and experiments of U-5.

**Table 9**
Comparison between the numerical and experimental results.

| Mesh type | Mesh size (cm) | Slab | Damaged area ($m^2$) | | | Damaged depth (cm) | | |
|---|---|---|---|---|---|---|---|---|
| | | | Experimental | Numerical | Error (%) | Experimental | Numerical | Error (%) |
| Unstructured | 1 | U-2 | 0.244 | 0.195 | 19.96 | 12.0 | 9 | 25.00 |
| | | U-4 | 0.213 | 0.243 | 13.92 | 11.7 | 13 | -11.11 |
| | | U-5 | 0.279 | 0.245 | 12.25 | 11.0 | 11 | 0 |
| | 2 | U-2 | 0.244 | 0.097 | 60.12 | 12.0 | 8 | 33.33 |
| | | U-4 | 0.213 | 0.125 | 41.18 | 11.7 | 10 | 14.53 |
| | | U-5 | 0.279 | 0.132 | 52.70 | 11.0 | 8 | 27.27 |
| | 5 | U-2 | 0.244 | 0.136 | 44.36 | 12.0 | 5 | 58.33 |
| | | U-4 | 0.213 | 0.136 | 35.96 | 11.7 | 10 | 14.53 |
| | | U-5 | 0.279 | 0.154 | 44.91 | 11.0 | 10 | 9.09 |
| Structured | 1 | U-2 | 0.244 | 0.166 | 31.97 | 12.0 | 8.0 | 33.33 |
| | | U-4 | 0.213 | 0.152 | 28.64 | 11.7 | 8.0 | 31.62 |
| | | U-5 | 0.279 | 0.159 | 43.01 | 11.0 | 9.0 | 18.18 |
| | 2 | U-2 | 0.244 | 0.150 | 38.52 | 12.0 | 4 | 66.67 |
| | | U-4 | 0.213 | 0.155 | 27.11 | 11.7 | 4 | 65.81 |
| | | U-5 | 0.279 | 0.226 | 18.86 | 11.0 | 6 | 45.45 |
| | 5 | U-2 | 0.244 | 0.133 | 45.40 | 12.0 | 5 | 58.33 |
| | | U-4 | 0.213 | 0.133 | 37.33 | 11.7 | 10 | 14.53 |
| | | U-5 | 0.279 | 0.182 | 34.84 | 11.0 | 15 | 9.09 |

Therefore, the number of unknown (vector) coefficients is in the summation can be computed as:

$$P = \binom{N+P}{P} = \frac{(N+P)!}{N!P!} \tag{19}$$

where the $N$ could be considered as the number of samples for imput parameters $n$. The PCE coefficients could fully determin a second-order random variable, namely that the output parameters of the computational model can be determined once the PCE coefficients are available. Therefore, an efficient procedures are essential to determin the PCE

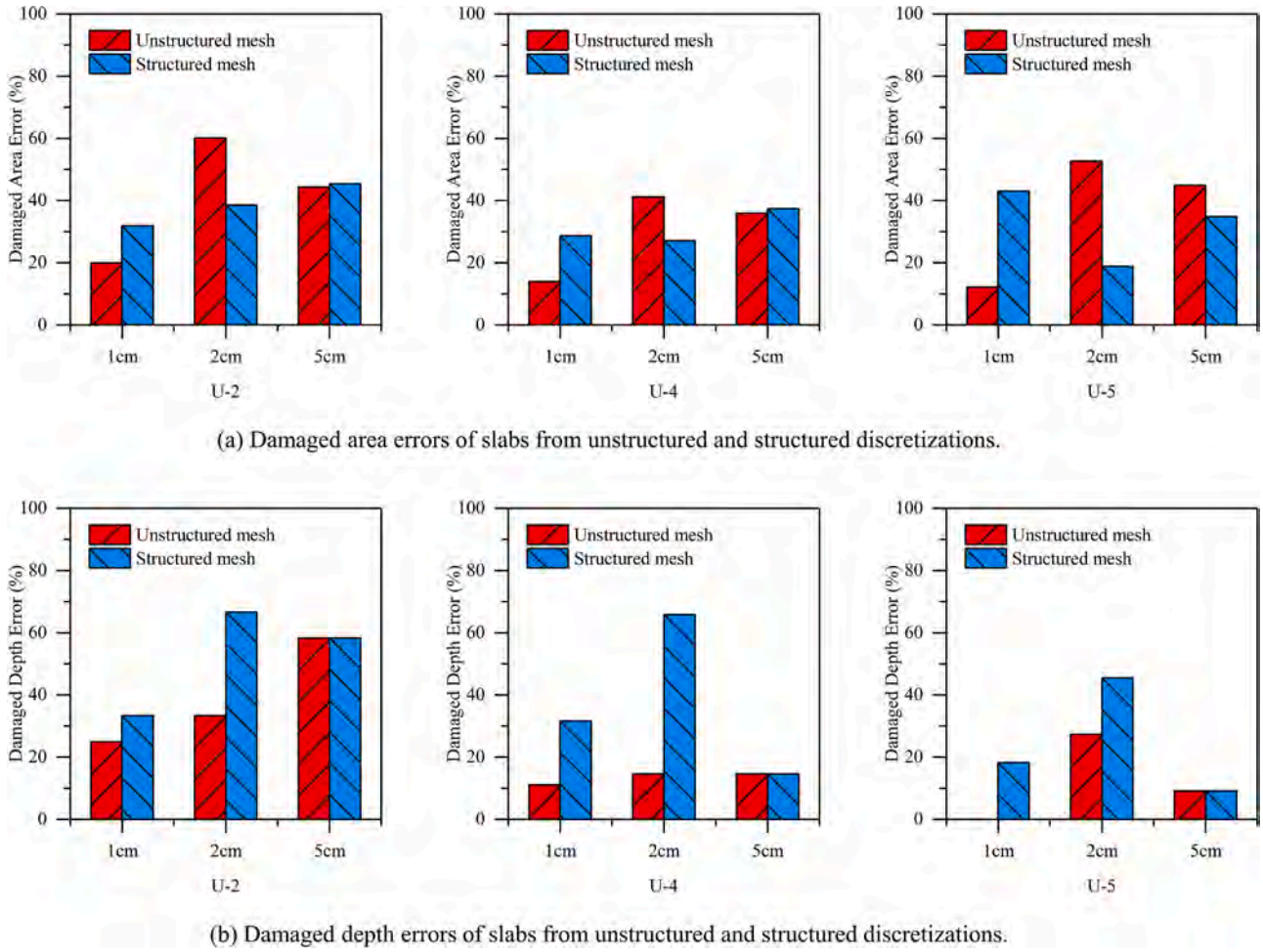


**Fig. 24.** Damaged area and depth errors of slabs from unstructured and structured discretizations.

coefficients of the output parameters **Y**. Classically, there are two methods for determining the PCE coefficients, i.e. the intrusive and non-intrusive methods. In the intrusive method, by means of a Galerkin projection of the model equations on the PCE basis, a weak solution of the mathematical model involving the random input is considered [51]. Whereas, this method requires modifications of the numerical code. In the non-intrusive method, it requires realizations of the deterministic code for the values of the input. There are two non-intrusive methods, including the least square approximation and the NISP [82]. In this study, the least square approximation method is employed.

Before determining the PCE coefficients, the experimental design $\mathbf{X} = \{\boldsymbol{X_1},\ldots,\boldsymbol{X_n}\}$ by choosing a samples with $n$ points firstly. To choose the samples, an appropriate sampling method should be used. In this study, the Latin Hypercube Sampling (LHS) [83]were employed in the probability space of the input parameters to obtain the samples. The LHS divide the space of $X_i$ into $n$ classes, which can obtain a reliable approximation of the stochastic properties of the random parameters.

Then transform the values **X** into standard variable $\boldsymbol{\xi}$ to obtain the matrix $\boldsymbol{\Psi}$. And put each point $\boldsymbol{X_i}$ in the generated sample into the computational model to obtain the corresponding output parameter $Y_i$, and generated a output vector $\boldsymbol{Y_s}$. Finally calculating the expansion coefficients. The sum of the squares of the residuals, $S$, between $\boldsymbol{Y_s}$ and the approximated solution of PCE, $\boldsymbol{Y_{as}}$,is as follows:

$$S = \sum_{s=1}^{n} \left( Y_s\left(\xi_1^s, \ldots, \xi_n^s\right) - \sum_{i=0}^{P-1} Z_j \Psi_j\left(\xi_1^s, \ldots, \xi_n^s\right) \right)^2 \tag{20}$$

where $\boldsymbol{\xi} = (\xi_1^s, \ldots, \xi_n^s)$is the transformed standard random vector of the sample point $\boldsymbol{X}$. By taking the partial derivative with respect to each of the coefficients equal to 0, $S$ can be minimized, i.e.

$$\frac{\partial S}{\partial Z_j} = 0; \quad j = 0, \ldots, P-1 \tag{21}$$

The PCE coefficients could be obtained by solving Eq. (21) and can be expressed in matrix notation as:

$$Z = \left(\psi^T \psi\right)^{-1} \psi^T Y \tag{22}$$

To solve the matrix, an appropriate number of samples ($N$) is essential. And $N$ can directly affect the results accuracy. On the one hand, a small $N$ could not obtain an effective surrogate model. On the other hand, a large number of $N$ increases the computational cost. Sudret [46] proposed an empirical rule on the optimal number of regression points, i.e. $N = (n\text{-}1)P$. The current study employed the empirical rule to obtain the samples. With the increase of the order of the polynomials, the number of unknown increases and requires larger number of samples. In the current study, the order of the polynomials was chosen as 3.

In this study, three input parameters were assumed being independent variables, i.e. the compressive strength of UHCTC, mass density of UHTCC, and maximum failure strain of UHTCC in the FE model. The FE model of UHTCC slab under 5 kg detonations (U-5) is employed as the computational model in this section. And the damaged area and depth, as well as the maximum deflection in the center of U-5 are considerd as the output parameters. The compressive strength related parameters were generated according to the K&C model use and validation [84], including the parameters of strength surfaces, elastic modulus and the EOS. The definitions and statistical properties of the uncertain input

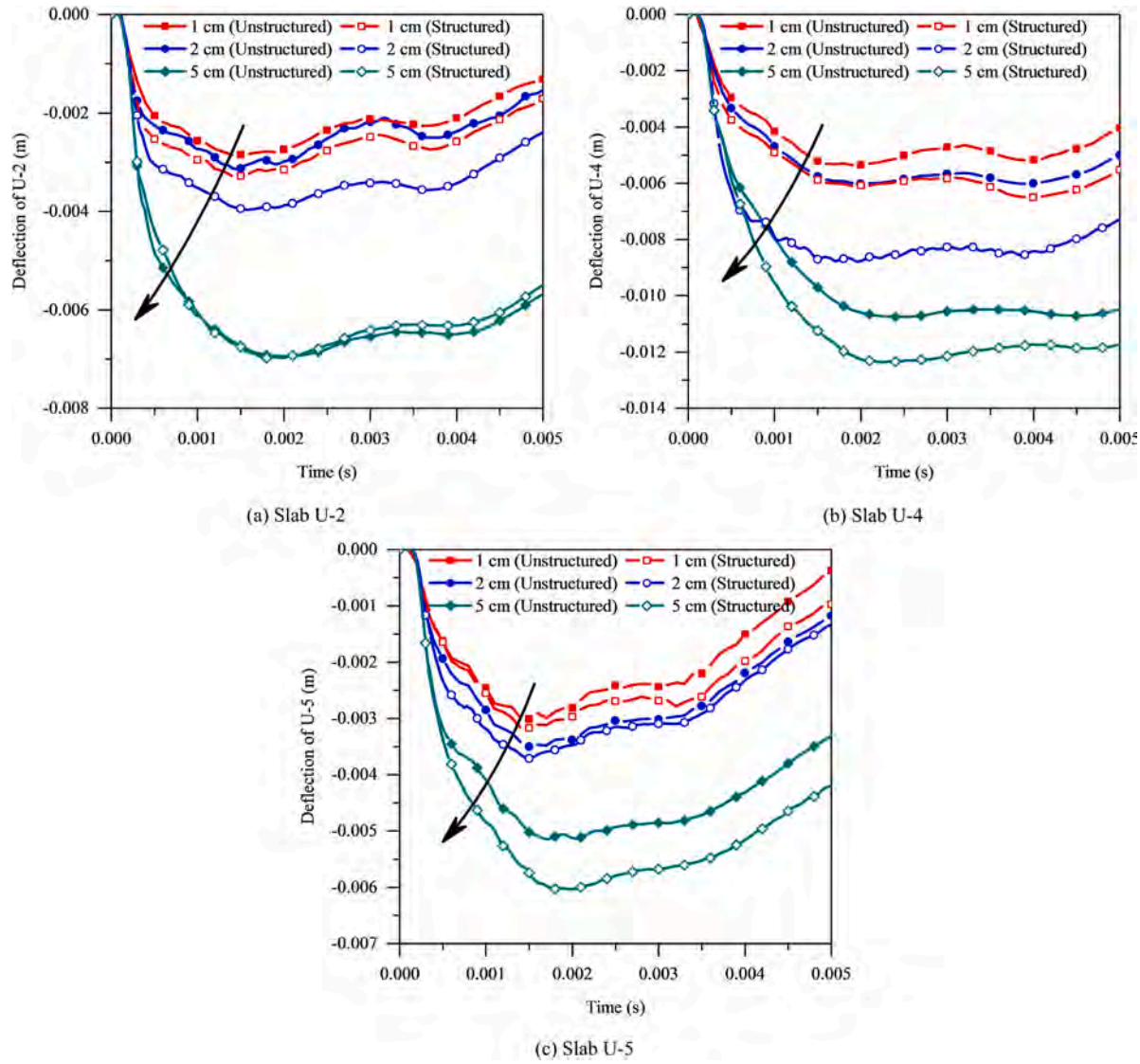


**Fig. 25.** Deflection-time cueves of slabs from structured and unstructured discretizations with different mesh size.

parameters are listed in Table 10. The normal distributions was used to describe the compressive strength of UHTCC [85]. The normal distributions was also used to describe the mass density of UHTCC, and the data range is 1800 kg/m$^3$ -2100 kg/m$^3$ [86]. The maximum failure strains are assumed as uniform distributions, and the data range is 0.005-0.08.

In this study, the coefficient of determination ($R^2$) is calculated. And the adjust coefficient of determination ($R^2_{adj}$) is also calculated. More details about the calculation of $R^2_{adj}$ are in Ref. [50]. The calculated value of $R^2$ and $R^2_{adj}$ for the constructed PCE are listed in Table 11. It is clear that the $R^2$ and $R^2_{adj}$ for the damaged area and maximum deflection are closed to 1.0, which means the PCE model can predict the results well. However, both the $R^2$ and $R^2_{adj}$ for the damaged depth are smaller values, that means the PCE model could not predict the damaged depth. It because the damaged depth in the numerical simulations is calculated according to the erosion of the element, which mainly depends on the mesh size. The damaged depth with the mesh size is a multiple relationship, namely that the value of the damaged depth jumps with the mesh size and could not be described using a continuous function.

### 6.2. Sensitivity analysis

The damaged area, damaged depth and the maximum deflection in the center of the slab are thought to be a random variable as a result of the randomness in the three output parameters. To quantify the sensitivity of the three input parameters on the output parameters, the Sobol's method was employed. The Sobol' indices can also be calculated based on the PCE. Based on variance decomposition of the output parameters, for an integrable function $f_{pc}(x)$ with $k$-dimentional unit typercube, the Sobol's function can be written as:

$$f_{pc}(x) = f_0 + \sum f_i + \sum_{i=0} \sum_{j=i+1} f_{ij} + \cdots + f_{12\ldots k} \tag{23}$$

By integrating the square of Eq. (23), it is possible to decompose the total variance as follows:

$$D = \sum_{i=1}^{n} D_i + \sum_{1 \le i \le j \le k} D_{ij} + \cdots + D_{12\ldots k} \tag{24}$$

The Sobol' indices are defined as follows:

$$S_{i_1,\ldots,i_s} = D_{i_1,\ldots,i_s} \big/ D \tag{25}$$

The Sobol' indices can be ontained from Eq. (24) devided by $D$.

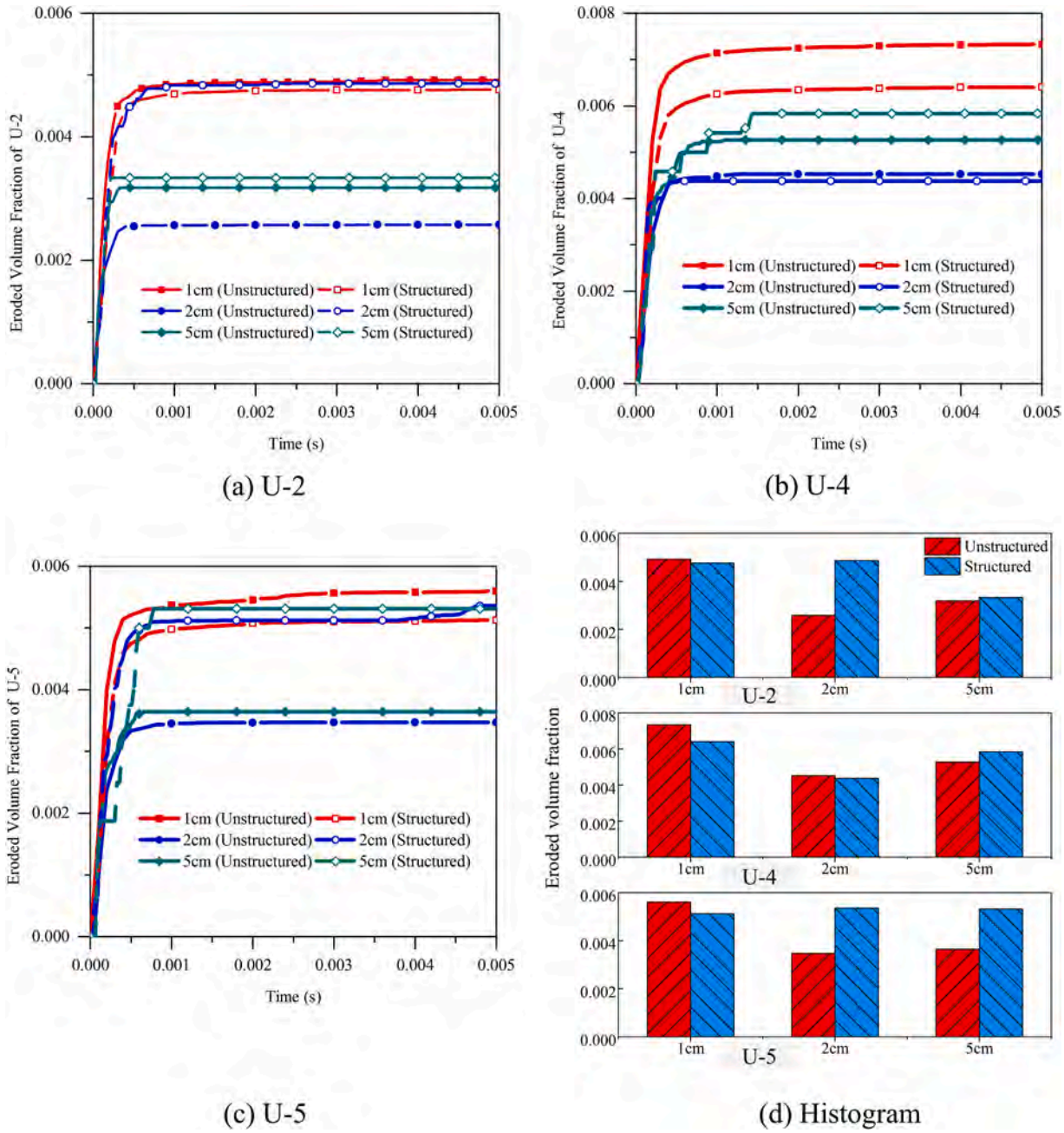


**Fig. 26.** Eroded volume fraction obtained from structured and unstructured discretizations with different mesh size.

**Table 10**
The definitions and statistical properties of the uncertain input parameters.

| Parameter definition | Unit | Symbol | Distribution type | Mean | Stantard deviation | Data range |
|---|---|---|---|---|---|---|
| Compressive strength | MPa | $f_c$ | Normal | 48 | 2.8 | (30,60) |
| Mass density | kg/m$^3$ | $\rho_u$ | Normal | 1950 | 30 | (1800,2100) |
| Maximum failure strain | [-] | $\varepsilon_{fmax}$ | Uniform | - | - | (0.005,0.08) |

$$\sum_{i=1}^{n} S_i + \sum_{1\le i\le j\le k} S_{ij} + \cdots + S_{12\ldots k} = 1 \tag{26}$$

where $S_i$ is the first order indices, which measures the influence of each parameters taken alone. And the higher order indices stand for possible mixed influence of the parameters. The total sensitivity indices $S_{Ti}$ is defined as follows to evaluate the contribution of each input parameter to the output parameters.

$$S_{T_i} = 1 - S_{-i} \tag{27}$$

where $S_{-i}$ is the sum of all $S_{i_1,\ldots,i_s}$ except index $i$.

The Sobol' indices are usually calculated by using Monte Carlo simulations. Whereas, the evaluation of $2^n$ Monte Carlo integrals are required to obtain a full description, which is not practically feasible unless $n$ is low. The first order indices can be obtained by the FAST method [87,88]. However the higher order indices increase the computational cost by Monte Carlo simulations. In addition, The Sobol'

**Table 11**
Properties of constructed PCEs.

| | Output parameters | | |
|---|---|---|---|
| | Damaged area | Deflection | Damaged depth |
| Coefficient of determination, $R^2$ | 0.9907 | 0.9972 | 0.5232 |
| Adjusted coefficient of determination, $R^2_{adj}$ | 0.9809 | 0.9943 | 0.0213 |

**Table 12**
Sobol' sensitivity indices based on the third order PCE, $n = 3$

| (a) Damaged area as the output parameter | | | | |
|---|---|---|---|---|
| Output parameter | Sobol' indices | Input parameter | | |
| Damaged area | | $f_c$ | $\rho_u$ | $\varepsilon_{fmax}$ |
| | $S_i$ | 0.0023 | 0.0514 | 0.8396 |
| | $\sum S_i = 0.8932$ | | | |
| | $S_{ij}$ | $S_{1j}$ | 0.0476 | 0.0142 |
| | | $S_{2j}$ | | 0.0449 |
| | $\sum S_{ij} = 0.1068$ | | | |
| | $S_{Ti}$ | 0.0641 | 0.1439 | 0.8987 |
| (b) The maximum deflection ($\delta_m$) of slab as the output parameter | | | | |
| Output parameter | Sobol' indices | Input parameter | | |
| $\delta_m$ | | $f_c$ | $\rho_u$ | $\varepsilon_{fmax}$ |
| | $S_i$ | 0.0088 | 0.1408 | 0.8090 |
| | $\sum S_i = 0.9586$ | | | |
| | $S_{ij}$ | $S_{1J}$ | 0.0221 | 0.0070 |
| | | $S_{2J}$ | | 0.0122 |
| | $\sum S_{ij} = 0.0414$ | | | |
| | $S_{Ti}$ | 0.0379 | 0.1752 | 0.8283 |
| (c) The damaged depth of slab as the output parameter | | | | |
| Output parameter | Sobol' indices | Input parameter | | |
| Damaged depth | | $f_c$ | $\rho_u$ | $\varepsilon_{fmax}$ |
| | $S_i$ | 0.0558 | 0.4283 | 0.1325 |
| | $\sum S_i = 0.6167$ | | | |
| | $S_{ij}$ | $S_{1j}$ | 0.1482 | 0.0219 |
| | | $S_{2j}$ | | 0.2132 |
| | $\sum S_{ij} = 0.3833$ | | | |
| | $S_{Ti}$ | 0.2259 | 0.7898 | 0.3677 |

indices can be calculated from the expansion coefficients of the PCE [46, 89] and the total variance of the response can be computed by:

$$\widehat{V} = \sum_{j=1}^{P-1} Z_j^2 E\left[\psi_j^2(X)\right] \tag{28}$$

where $E$ [-] is the mathematical expectation. The Sobol' indices can be calculated by assembling, square summing and subsequently normalizing the expansion coefficients. The Sobol' sensitivity based on the third order PCE are listed in Table 12. And the total effect Sobol' sensitivity indices for the three output parameters are displayed in Fig. 27. For the damaged area, the effect of the maximum failure strain is 83.96%, that means the damaged area mainly depends on the maximum failure strain. That because the damaged area is under plastic flow state under detonations, where the failure of the element depends on the errosions. For the maximum deflection, the effect of the maximum strain is 80.90%. That because the maximum failure strain controls the failure of the elements, and the failure of elements affects the load transferring. For the damaged depth, the mass density, the mass density and the maximum failure strain contribute the effect in sequence.

## 7. Conclusion and future work

### 7.1. Conclusion

This study conducts the experiment, numerical simulations and the uncertainty analysis researches on the UHTCC under detonation. In the experiment, three UHTCC slabs and three ordinary concrete slabs were prepared. 2 kg, 4 kg and 5 kg TNT explosives were employed to generate the blast loadings. The cratering areas, cratering depths, and spalling areas of all slabs were quantitatively recorded, analyzed, and compared. The existing empirical methods were utilized for predicting the local damage mode of slabs and were employed to explore the feasibility of the empirical methods for UHTCC. In the numerical simulations, the structured and unstructured discretizations were used in the FE model. And the numerical results from the two types of the discretizations were compared. The effect of mesh size on the damaged area, damaged depth, deflection-time curves of the slab and the eroded volume fraction were compared and discussed. In the uncertain analysis, the PCE model and Sobol sensitivity were used. The maximum failure strain, the compressive strength and the mass density of UHTCC were employed as the input parameters to investigate and discuss their effect on the value of output parameters, which include the damaged area, damaged depth and maximum deflection of UHTCC slab. The following conclusions can be drawn:

(1) In the blast experiment, cratering and severe spalling damage appeared in ordinary concrete slabs (NC-2, NC-4 and NC-5). In the top surface of UHTCC slabs, cratering damage also appeared. However, in the bottom surface, swelling damage, spalling damage and no damage appeared in U-2, U-4 and U-5, respectively. The UHTCC slabs presented significantly decrease the spalling damage than the concrete slabs. The existing empirical prediction methods were employed to evaluate the damage mode of slabs in this study. And McVay's [60] and Morishita's [6] prediction methods give reasonable predictions on concrete slabs but significantly underestimate the spalling resistance of UHTCC slabs. In general, the findings of this study indicate that the UHTCC slabs have favorable performance under blast loading and have potential in the construction of fortifications.

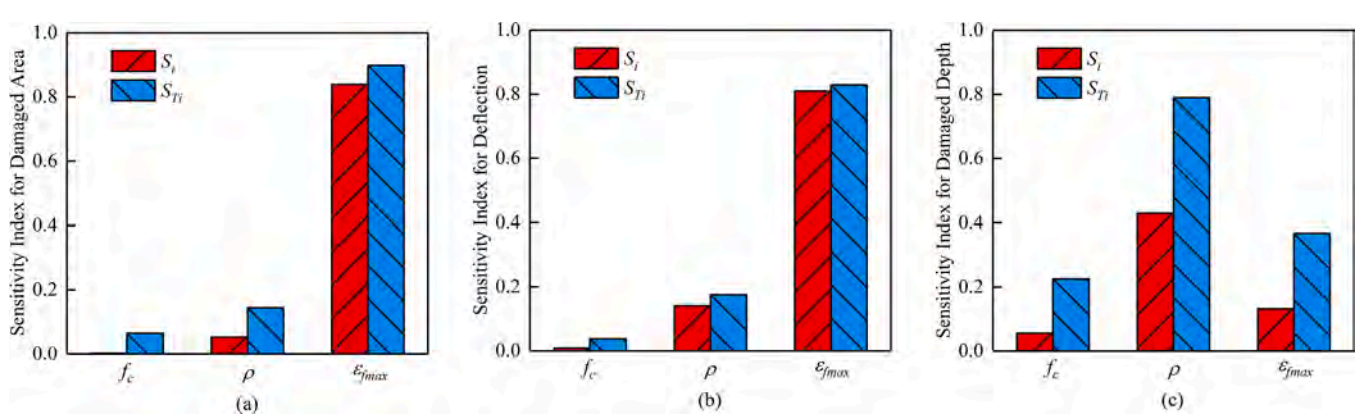


**Fig. 27.** The total-effect Sobol' sensitivity indices for three output parameters.

(2) In the numerical simulations, when the mesh size is 10 mm, the results from the unstructured discretization shows better agreement with the experimental results, which proved that the unstructured discretizations can provide accurate prediction in the blast simulations. When the mesh size increases from 10 mm to 50 mm, the deflection increases, and the maximum deflection of slab from the unstructured discretizations presents smaller than that from the structured discretizations. The eroded volume fraction of slabs decrease with the unstructured discretizations from 10 mm to 20 mm while increase from 20 mm to 50 mm, while that from the structured discretizations show unconspicuous patterns.

(3) In the uncertain analysis, The $R^2$ and $R^2_{adj}$ for the damaged area and maximum deflection are closed to 1.0. That means the PCE model can predict the results well. But both the $R^2$ and $R^2_{adj}$ for the damaged depth are smaller values, that means the PCE model could not predict the damaged depth. It because the value of the damaged depth jumps with the mesh size and could not be described using a continuous function. From the global sensibility analysis, the maximum failure strain mainly provide influence on the damaged area and the maximum deflection of slab, and the compressive strength provides the smallest influence on the damaged area and maximum deflection of slab. The damaged area and the maximum deflection of slab mainly depend on the maximum failure strain, and the mass density. For the damaged depth, the compressive strength, the mass density and the maximum failure strain all contribute the effect.

### 7.2. Future work

Although a few works were conducted in this study, there are still some works need to be researched in the future. In the constitutive model for UHTCC, the constitutive model for UHTCC will be further modified according to dynamic material tests to obtain more accurate predictions on the dynamic performance of UHTCC. In the FE model for blast simulations, some ideal sophisticated numerical methods, such as particle-based methods [30–32], DH-PD [33,34], NHPD [35], CEMs [36–38], PFM [39] should be learned and employed in the FE model.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

## Acknowledgements

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## References

[1] Li J, Hao H. Numerical study of concrete spall damage to blast loads. Int J Impact Eng 2014;68:41–55. https://doi.org/10.1016/j.ijimpeng.2014.02.001.
[2] Gebbeken N, Ruppert M. A new material model for concrete in high-dynamic hydrocode simulations. Arch Appl Mech 2000;70:463–78. https://doi.org/10.1007/s004190000079.
[3] Morris WR, Haik BG, Osborn FD, Fleming JC. Ocular injuries sustained by survivors of the Oklahoma City bombing. Ophthalmology 2000;107:837–43. https://doi.org/10.1016/S0161-6420(00)00030-0.
[4] Mallonee S, Shariat S, Stennies G, Waxweiler R, Hogan D, Jordan F. Physical injuries and fatalities resulting from the Oklahoma City bombing. J Am Med Assoc 1996;276:382–7. https://doi.org/10.1001/jama.276.5.382.
[5] Brismar B, Bergenwald L. The terrorist bomb explosion in Bologna, Italy, 1980: An analysis of the effects and injuries sustained. J Trauma - Inj Infect Crit Care 1982; 22:216–20. https://doi.org/10.1097/00005373-198203000-00007.
[6] Morishita M, Tanaka H, Ando T, Hagiya H. Effects of Concrete Strength and Reinforcing Clear Distance on the Damage of Reinforced Concrete Slabs Subjected to Contact Detonations. Concr Res Technol 2004;15:89–98. https://doi.org/10.3151/crt1990.15.2_89.
[7] TANAKA H, TUJI M. Effects of Reinforcing on Damage of Reinforced Concrete Slabs subjected to Explosive loading. Concr Res Technol 2003;14:1–11. https://doi.org/10.3151/crt1990.14.1_1.
[8] Nam J, Kim H, Kim G. Experimental Investigation on the Blast Resistance of Fiber-Reinforced Cementitious Composite Panels Subjected to Contact Explosions. Int J Concr Struct Mater 2017;11:29–43. https://doi.org/10.1007/s40069-016-0179-y.
[9] Li VC, Leung CKY. Steady state and multiple cracking of short random fiber composites. J Eng Mech 1992;118:2246–64.
[10] Leung Christopher KY. DESIGN CRITERIA FOR PSEUDODUCTILE FIBER-REINFORCED COMPOSITES. J Eng Mech 1996;122:10–8.
[11] Li Q, Zhao X, Xu S, Leung CKY, Wang B. Multiple Impact Resistance of Hybrid Fiber Ultrahigh Toughness Cementitious Composites with Different Degrees of Initial Damage. J Mater Civ Eng 2019;31:04018368. https://doi.org/10.1061/(asce)mt.1943-5533.0002576.
[12] Li H, Xu S, Leung CKY. Tensile and flexural properties of ultra high toughness cemontious composite. J Wuhan Univ Technol Mater Sci Ed 2009;24:677–83. https://doi.org/10.1007/s11595-009-4677-5.
[13] Zhang D, Yu J, Wu H, Jaworska B, Ellis BR, Li VC. Discontinuous micro-fibers as intrinsic reinforcement for ductile Engineered Cementitious Composites (ECC). Compos Part B Eng 2020;184:107741. https://doi.org/10.1016/j.compositesb.2020.107741.
[14] Li H, Xu S. Determination of energy consumption in the fracture plane of Ultra High Toughness Cementitious Composite with direct tension test. Eng Fract Mech 2011;78. https://doi.org/10.1016/j.engfracmech.2011.03.012.
[15] Roesler J, Paulino GH, Park K, Gaedicke C. Concrete fracture prediction using bilinear softening. Cem Concr Compos 2007;29. https://doi.org/10.1016/j.cemconcomp.2006.12.002.
[16] Mohamed Maalej VCL. Introduction of Strain-Hardening Engineered Cementitious Composites in Design of Reinforced Concrete Flexural Members for Improved Durability. ACI Struct J 1995;92:167–76. https://doi.org/10.14359/1150.
[17] Li Q-H, Sun C-J, Lyu J-F, Quan G, Huang B-T, Xu S-L. Fire Performance of Steel-Reinforced Ultrahigh-Toughness Cementitious Composite Columns: Experimental Investigation and Numerical Analyses. J Struct Eng 2020;146:04020012. https://doi.org/10.1061/(asce)st.1943-541x.0002567.
[18] Li QH, Sun CJ, Xu SL. Thermal and mechanical properties of ultrahigh toughness cementitious composite with hybrid PVA and steel fibers at elevated temperatures. Compos Part B Eng 2019;176:107201. https://doi.org/10.1016/j.compositesb.2019.107201.
[19] Lu C, Yu J, Leung CKY. Tensile performance and impact resistance of Strain Hardening Cementitious Composites (SHCC) with recycled fibers. Constr Build Mater 2018;171:566–76. https://doi.org/10.1016/j.conbuildmat.2018.03.108.
[20] Li Q, Huang B, Xu S, Zhou B, Yu RC. Compressive fatigue damage and failure mechanism of fiber reinforced cementitious material with high ductility. Cem Concr Res 2016;90:174–83. https://doi.org/10.1016/j.cemconres.2016.09.019.
[21] Huang BT, Li QH, Xu SL, Zhang L. Static and fatigue performance of reinforced concrete beam strengthened with strain-hardening fiber-reinforced cementitious composite. Eng Struct 2019;199:109576. https://doi.org/10.1016/j.engstruct.2019.109576.
[22] Huang BT, Li QH, Xu SL, Liu W, Wang HT. Fatigue deformation behavior and fiber failure mechanism of ultra-high toughness cementitious composites in compression. Mater Des 2018;157:457–68. https://doi.org/10.1016/j.matdes.2018.08.002.
[23] Zhao X, Li Q, Xu S. Contribution of steel fiber on the dynamic tensile properties of hybrid fiber ultra high toughness cementitious composites using Brazilian test. Constr Build Mater 2020;246:118416. https://doi.org/10.1016/j.conbuildmat.2020.118416.
[24] Yang EH, Li VC. Tailoring engineered cementitious composites for impact resistance. Cem Concr Res 2012;42:1066–71. https://doi.org/10.1016/j.cemconres.2012.04.006.
[25] Zhang J, Maalej M, Quek ST. Performance of Hybrid-Fiber ECC Blast/Shelter Panels Subjected to Drop Weight Impact. J Mater Civ Eng 2007;19:855–63. https://doi.org/10.1061/(asce)0899-1561(2007)19:10(855).
[26] Li QH, Huang BT, Xu SL. Development of assembled permanent formwork using ultra high toughness cementitious composites. Adv Struct Eng 2016;19:1142–52. https://doi.org/10.1177/1369433216634495.
[27] Huang BT, Li QH, Xu SL, Li CF. Development of reinforced ultra-high toughness cementitious composite permanent formwork: Experimental study and Digital Image Correlation analysis. Compos Struct 2017;180:892–903. https://doi.org/10.1016/j.compstruct.2017.08.016.
[28] Huang BT, Li QH, Xu SL, Zhou B. Strengthening of reinforced concrete structure using sprayable fiber-reinforced cementitious composites with high ductility. Compos Struct 2019;220:940–52. https://doi.org/10.1016/j.compstruct.2019.04.061.
[29] Li VC, Bos FP, Yu K, McGee W, Ng TY, Figueiredo SC, et al. On the emergence of 3D printable Engineered, Strain Hardening Cementitious Composites (ECC/SHCC). Cem Concr Res 2020;132:106038. https://doi.org/10.1016/j.cemconres.2020.106038.
[30] Rabczuk T, Belytschko T. Cracking particles: A simplified meshfree method for arbitrary evolving cracks. Int J Numer Methods Eng 2004;61:2316–43. https://doi.org/10.1002/nme.1151.

[31] Rabczuk T, Zi G, Bordas S, Nguyen-Xuan H. A simple and robust three-dimensional cracking-particle method without enrichment. Comput Methods Appl Mech Eng 2010;199:2437–55. https://doi.org/10.1016/j.cma.2010.03.031.
[32] Rabczuk T, Belytschko T. A three-dimensional large deformation meshfree method for arbitrary evolving cracks. Comput Methods Appl Mech Eng 2007;196:2777–99. https://doi.org/10.1016/j.cma.2006.06.020.
[33] Ren Huilong, Zhuang Xiaoying. Yongchang Cai TR. Dual-horizon peridynamics. Int J Numer Methods Eng 2016;108:1451–67. https://doi.org/10.1002/nme.5257.
[34] Ren H, Zhuang X, Rabczuk T. Dual-horizon peridynamics: A stable solution to varying horizons. Comput Methods Appl Mech Eng 2017;318:762–82. https://doi.org/10.1016/j.cma.2016.12.031.
[35] Zhang Y, Yang X, Wang X, Zhuang X. A micropolar peridynamic model with non-uniform horizon for static damage of solids considering different nonlocal enhancements. Theor Appl Fract Mech 2021;113:102930. https://doi.org/10.1016/j.tafmec.2021.102930.
[36] Zhang Y, Zhuang X. Cracking elements method for dynamic brittle fracture. Theor Appl Fract Mech 2019;102:1–9. https://doi.org/10.1016/j.tafmec.2018.09.015.
[37] Zhang Y, Zhuang X. Cracking elements: A self-propagating Strong Discontinuity embedded Approach for quasi-brittle fracture. Finite Elem Anal Des 2018;144:84–100. https://doi.org/10.1016/j.finel.2017.10.007.
[38] Zhang Y, Mang HA. Global cracking elements: A novel tool for Galerkin-based approaches simulating quasi-brittle fracture. Int J Numer Methods Eng 2020;121:2462–80. https://doi.org/10.1002/nme.6315.
[39] Zhou S, Zhuang X, Zhu H, Rabczuk T. Phase field modelling of crack propagation, branching and coalescence in rocks. Theor Appl Fract Mech 2018;96:174–92. https://doi.org/10.1016/j.tafmec.2018.04.011.
[40] Rabczuk Timon, Gracie Robert. Jeong-Hoon Song TB. Immersed particle method for fluid–structure interaction. Int J Numer Methods Eng 2010;81:48–71. https://doi.org/10.1002/nme.2670.
[41] Koobus B, Farhat C. Second-order time-accurate and geometrically conservative implicit schemes for flow computations on unstructured dynamic meshes. Comput Methods Appl Mech Eng 1999;170:103–29. https://doi.org/10.1016/S0045-7825(98)00207-2.
[42] Tsoutsanis P, Kokkinakis IW, Könözsy L, Drikakis D, Williams RJR, Youngs DL. Comparison of structured- and unstructured-grid, compressible and incompressible methods using the vortex pairing problem. Comput Methods Appl Mech Eng 2015;293:207–31. https://doi.org/10.1016/j.cma.2015.04.010.
[43] Mavriplis DJ. Unstructured-mesh discretizations and solvers for computational aerodynamics. AIAA J 2008;46:1281–98. https://doi.org/10.2514/1.34681.
[44] McGrory WD, Walters RW, Lohner R. Three-dimensional space-marching algorithm on unstructured grids. AIAA J 1991;29:1844–9. https://doi.org/10.2514/3.10809.
[45] Xiu D, Karniadakis GE. Modeling uncertainty in flow simulations via generalized polynomial chaos. J Comput Phys 2003;187:137–67. https://doi.org/10.1016/S0021-9991(03)00092-5.
[46] Sudret B. Global sensitivity analysis using polynomial chaos expansions. Reliab Eng Syst Saf 2008;93:964–79. https://doi.org/10.1016/j.ress.2007.04.002.
[47] Crestaux T, Le Maître O, Martinez JM. Polynomial chaos expansion for sensitivity analysis. Reliab Eng Syst Saf 2009;94:1161–72. https://doi.org/10.1016/j.ress.2008.10.008.
[48] Hamdia KM, Ghasemi H, Zhuang X, Alajlan N, Rabczuk T. Sensitivity and uncertainty analysis for flexoelectric nanostructures. Comput Methods Appl Mech Eng 2018;337:95–109. https://doi.org/10.1016/j.cma.2018.03.016.
[49] Hamdia KM, Msekh MA, Silani M, Thai TQ, Budarapu PR, Rabczuk T. Assessment of computational fracture models using Bayesian method. Eng Fract Mech 2019;205:387–98. https://doi.org/10.1016/j.engfracmech.2018.09.019.
[50] Hamdia KM, Silani M, Zhuang X, He P, Rabczuk T. Stochastic analysis of the fracture toughness of polymeric nanoparticle composites using polynomial chaos expansions. Int J Fract 2017;206:215–27. https://doi.org/10.1007/s10704-017-0210-6.
[51] Ghanem RG, Spanos PD. Stochastic Finite Elements: A Spectral Approach. Stoch Finite Elem A Spectr Approach 1991:7795. https://doi.org/10.1007/978-1-4612-3094-6.
[52] Li Q-H, Yin X, Huang B-T, Luo A-M, Lyu Y, Sun C-J, et al. Shear Interfacial Fracture of Strain-Hardening Fiber-Reinforced Cementitious Composites and Concrete: A Novel Approach. Eng Fract Mech 2021;253:107849. https://doi.org/10.1016/j.engfracmech.2021.107849.
[53] China National Standards JGJ/T 70-2009. Standard for test method of basic properties of construction mortar. China Architecture & Building Press; 2009.
[54] Japan Society of Civil Engineers. Recommendations for Design and Construction of High Performance Fiber Reinforced Cement Composites with Multiple Fine Cracks (HPFRCC). Concr Eng Ser 2008;82:1–16.
[55] Yao W, Sun W, Shi Z, Chen B, Chen L, Feng J. Blast-resistant performance of hybrid fiber-reinforced concrete (HFRC) panels subjected to contact detonation. Appl Sci 2020;10. https://doi.org/10.3390/app10010241.
[56] Kot CA, Valentin RA, McLennan DA, Turula P. Effects of air blast on power plant structures and components. Argonne Natl Lab; 1978.
[57] Kot C A. Spalling of concrete walls under blast load 1977.
[58] Li J, Hao H. Numerical and theoretical study of concrete spall damage under blast loads. Appl. Mech. Mater. 2014;553:774–9. https://doi.org/10.4028/www.scientific.net/AMM.553.774.
[59] DoD. Structures to resist the effects of accidental explosions, UFC 3-340-02, 298. Washington, DC, USA: US Department of Defense; 2008.
[60] Mcvay MK. Spall Damage of Concrete Structures. US Army Corps Eng 1988.
[61] Malvar LJ, Crawford JE, Wesevich JW, Simons D. A plasticity concrete material model for DYNA3D. Int J Impact Eng 1997;19:847–73. https://doi.org/10.1016/s0734-743x(97)00023-7.
[62] Ranade R, Li VC. Material model for simulating strain-hardening cementitious composites in LS-DYNA. In: SHCC3; 2014. p. 235–42.
[63] Xu Shilang, Wu Ping, Li Qinghua, Zhou Fei, Li Rui. Determination of K&C model parameters for ultra-high toughness cementitious composites. J Build Struct 2021. https://doi.org/10.14006/j.jzjgxb.2020.0587.
[64] Li J, Zhang YX. Evolution and calibration of a numerical model for modelling of hybrid-fibre ECC panels under high-velocity impact. Compos Struct 2011;93:2714–22. https://doi.org/10.1016/j.compstruct.2011.05.033.
[65] Wang W, Zhang D, Lu F, chuan Wang S, Tang F. Experimental study and numerical simulation of the damage mode of a square reinforced concrete slab under close-in explosion. Eng Fail Anal 2013;27:41–51. https://doi.org/10.1016/j.engfailanal.2012.07.010.
[66] Li Q, Zhao X, Xu S, Gao X. Influence of steel fiber on dynamic compressive behavior of hybrid fiber ultra high toughness cementitious composites at different strain rates. Constr Build Mater 2016;125:490–500. https://doi.org/10.1016/j.conbuildmat.2016.08.066.
[67] Dong Li H, Lang Xu S. Rate dependence of ultra high toughness cementitious composite under direct tension. J Zhejiang Univ Sci A 2016;17:417–26. https://doi.org/10.1631/jzus.A1600031.
[68] Yang EH, Li VC. Strain-rate effects on the tensile behavior of strain-hardening cementitious composites. Constr Build Mater 2014;52:96–104. https://doi.org/10.1016/j.conbuildmat.2013.11.013.
[69] Ranade R, Li VC, Heard WF. Tensile rate effects in High Strength-High Ductility Concrete. Cem Concr Res 2015;68:94–104. https://doi.org/10.1016/j.cemconres.2014.11.005.
[70] Tu Z, Lu Y. Evaluation of typical concrete material models used in hydrocodes for high dynamic response simulations. Int J Impact Eng 2009;36:132–46. https://doi.org/10.1016/j.ijimpeng.2007.12.010.
[71] Xu H, Wen HM. Semi-empirical equations for the dynamic strength enhancement of concrete-like materials. Int J Impact Eng 2013;60:76–81. https://doi.org/10.1016/j.ijimpeng.2013.04.005.
[72] Mechtcherine V, De Andrade Silva F, Butler M, Zhu D, Mobasher B, Gao SL, et al. Behaviour of strain-hardening cement-based composites under high strain rates. J Adv Concr Technol 2011;9:51–62. https://doi.org/10.3151/jact.9.51.
[73] Mechtcherine V, Silva FDA, Müller S, Jun P, Filho RDT. Coupled strain rate and temperature effects on the tensile behavior of strain-hardening cement-based composites (SHCC) with PVA fibers. Cem Concr Res 2012;42:1417–27. https://doi.org/10.1016/j.cemconres.2012.08.011.
[74] Maalej M, Quek ST, Zhang J. Behavior of Hybrid-Fiber Engineered Cementitious Composites Subjected to Dynamic Tensile Loading and Projectile Impact. J Mater Civ Eng 2005;17:143–52. https://doi.org/10.1061/(asce)0899-1561(2005)17:2(143.
[75] Boshoff WP, van Zijl GPAG. Time-dependent response of ECC: Characterisation of creep and rate dependence. Cem Concr Res 2007;37:725–34. https://doi.org/10.1016/j.cemconres.2007.02.001.
[76] Yu KQ, Dai JG, Lu ZD, Poon CS. Rate-dependent tensile properties of ultra-high performance engineered cementitious composites (UHP-ECC). Cem Concr Compos 2018;93:218–34. https://doi.org/10.1016/j.cemconcomp.2018.07.016.
[77] Xu S, Wu P, Zhou F, Jiang X, Chen B, Li Q. A dynamic constitutive model of ultra high toughness cementitious composites. J Zhejiang Univ A 2020;21:939–60. https://doi.org/10.1631/jzus.a1900599.
[78] LSTC. LS-DYNA Keyword User's Manual. CA: Livermore Software Technology Corporation (LSTC); 2014.
[79] Tabatabaei ZS, Volz JS, Baird J, Gliha BP, Keener DI. Experimental and numerical analyses of long carbon fiber reinforced concrete panels exposed to blast loading. Int J Impact Eng 2013;57:70–80. https://doi.org/10.1016/j.ijimpeng.2013.01.006.
[80] Hamdia KM, Marino M, Zhuang X, Wriggers P, Rabczuk T. Sensitivity analysis for the mechanics of tendons and ligaments: Investigation on the effects of collagen structural properties via a multiscale modeling approach. Int j Numer Method Biomed Eng 2019;35:1–19. https://doi.org/10.1002/cnm.3209.
[81] Wiener NW. The Homogeneous Chaos. Am J Math 1938;60:897–936. https://doi.org/10.2307/2371268.
[82] Reagan Matthew T, Najm Habib N, Ghanem Roger G, Knio Omar M. Uncertainty quantification in reacting-flow simulations through non-intrusive spectral projection. Combustion and Flame 2003;132:545–55. https://doi.org/10.1016/S0010-2180(02)00503-5.
[83] Conover Rlij. A distribution-free approach to inducing rank correlation among input variables. Commun Stat - Simul Comput 1982;11:311–34.
[84] Karagozian & Case . USE AND VALIDATION OF THE RELEASE III K&C CONCRETE MATERIAL MODEL IN LS-DYNA. 2011.
[85] Tumidajski P J, Fiore L, Khodabocus T, et al. Comparison of Weibull and normal distributions for concrete compressive strengths. Can J Civ Eng 2006;33:1287–92. https://doi.org/10.1139/L06-080.
[86] Thomas RJ, Sorensen AD. Review of strain rate effects for UHPC in tension. Constr Build Mater 2017;153:846–56. https://doi.org/10.1016/j.conbuildmat.2017.07.168.
[87] Cukier RI, Levine HB, Shuler KE. Nonlinear sensitivity analysis of multiparameter model systems. J Comput Phys 1978;26:1–42.
[88] Vu-Bac N, Rafiee R, Zhuang X, Lahmer T, Rabczuk T. Uncertainty quantification for multiscale modeling of polymer nanocomposites with correlated parameters. Compos Part B Eng 2015;68:446–64. https://doi.org/10.1016/j.compositesb.2014.09.008.
[89] Crestaux T, Matre O L MJM. Polynomial chaos expansion for sensitivity analysis. Reliab Eng Syst Saf 2009;94:1161–72. https://doi.org/10.1016/j.ress.2008.10.008.